AD 2. AERODROMES

VOMM AD 2.1 AERODROME LOCATION INDICATOR AND NAME

VOMM - CHENNAI INTERNATIONAL AIRPORT, CHENNAI

VOMM AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

1	Aerodrome reference point coordinates and its site	125942N 0801025E 272 DEG/527M from intersection of RWY 07/25 and 12/30.	
2	Direction and distance of aerodrome reference point from the center of the city or town which the aerodrome serves	228 DEG/17KM from central railway station.	
3	Aerodrome elevation and reference temperature	54 FT / 38.0 DEG C	
4	Magnetic variation, date of information and annual change	1.75 DEG W (2010) /0.017 DEG E	
5	Name of aerodrome operator, address, telephone, telefax, e-mail address, AFS address, website (if available)	The Airport Director, Airports Authority of India, Administrative Block, Chennai Airport, Chennai -600027,	
		Telephone:	(91) 44 22561122 (91) 44 22561234
		Fax:	(91) 44 22560512 (91) 44 22561010
		AFS:	VOMMATYX, VOMMYUYU
		Email:	apdchennai@aai.aero
6	Types of traffic permitted (IFR/VFR)	IFR/VFR	
7	Remarks	i). Non-scheduled aircraft operators are required to file their slot requests with the Airport Operations Control Centre (AOCC) for clearance. All non-scheduled and General Aviation movements are subject to approval from the AOCC. ii). Aerodrome Reference Code: 4E. iii). Freighter Aircraft (B747-800F & AN124 (Code -F) will be accommodated on RWY 07/25 subject to 24 Hrs. prior coordination and approval. iv). Gravity Model: EGM-08	

VOMM AD 2.3 OPERATIONAL HOURS

1	Aerodrome Operator	MON-FRI 0400-1230 UTC (0930-1800 IST) SAT, SUN+HOL: NIL
2	Custom and immigration	H24
3	Health and sanitation	H24
4	AIS briefing office	H24
5	ATS reporting office (ARO)	H24
6	MET Briefing office	H24
7	Air Traffic Service	H24
8	Fuelling	H24
9	Handling	H24
10	Security	H24
11	De-icing	NIL

12	Remarks	(i) Aerodrome available for all weather operations (AWO). Day and night for all flights. (ii) RWY 12/30 available for all weather operations (AWO) for ACFT upto CODE C for landing & take-off and up to code F for towing/taxiing. iii. The Approved Hourly Runway Traffic Handling Capacity As Following: 1. Peak Capacity When Rwy In Use 07/25 Max No. of Arrival and Departure-36 Max No. of Arrival Only -26 Max No. of Departure Only -27 2. When Secondary Runway 12/30 In Use For Code C Ops Max No. of Arrival and Departure-36 Max No. of Arrival Only -26 Max No. of Departure Only -27

VOMM AD 2.4 HANDLING SERVICES AND FACILITIES

1	Cargo-handling facilities	International Annual Handling capacity: 12,18,608 MT Domestic Annual Handling capacity: 2,09,072 MT
2	Fuel and Oil types	JET A1 Jet A1 NIL
3	Fuelling facilities and capacity	1. Indian Oil Corp.: i. Two Nrs of Refuellers of 16,000 Litres capacity with discharge rate of 15 to 20 Litres / second. ii. One Nr of Refueller of 11,000 Litres capacity with discharge rate of 15 to 20 Litres / second. iii. Two Nrs of Refuellers of 27,000 Litres capacity with discharge rate of 40 Litres / second. iv. Four Nrs of Refuellers of 45,000 Litres capacity with discharge rate of 40 to 50 Litres / second. 2. Hindustan Petroleum Corp Ltd.: i. Three Nr of Refueller of 25,000 Litres capacity with discharge rate of 40 Litres / second. ii. Three Nr of Refueller of 15,000 Litres capacity with discharge rate of 15 Litres / second. iii. One Nr of Refueller of 6,000 Litres capacity with discharge rate of 6 Litres / second. iv. One Nr of Refueller of 27,000 Litres capacity with discharge rate of 40 Litres / second. 3. Bharat Petroleum Corp Ltd.: i. One Nr of Refueller of 45,000 Litres capacity with discharge rate of 40 Litres / second. ii. Two Nr of Refueller of 15,000 Litres capacity with discharge rate of 20 to 25 Litres / second. iii. One Nr of Refueller of 23,000 Litres capacity with discharge rate of 30 Litres / second. iv. One Nr of Refueller of 21,900 Litres capacity with discharge rate of 30 Litres / second. v. Two Nr of Refueller of 25,000 Litres capacity with discharge rate of 40 Litres / second. vi. One Nr of Refueller of 36,000 Litres capacity with discharge rate of 40 Litres / second. vii. One Nr of Refueller of 40,000 Litres capacity with discharge rate of 40 Litres / second.
4	De-icing facilities	NIL

5	Hangar space for visiting aircraft	AAI & AIESL Hanger subject to availability
6	Repair facilities for visiting aircraft	Available by arrangement with AIESL
7	Remarks	Following Ground Handlers are available at Chennai Airport i. AI Airport Services Ltd. ii. Bird Airport Services (Chennai) Pvt Ltd.

VOMM AD 2.5 PASSENGER FACILITIES

1	Hotel(s) at or in the vicinity of aerodrome	Near the AD and in the city.
2	Restaurant(s) at or in the vicinity of aerodrome	At AD and in the city
3	Transportation possibilities	Metro,Buses,taxis and car hire from AD.Trains to and from city.
4	Medical Facilities	First aid at AD. Nursing home and Hospitals near AD and in the city.
5	Bank and post office at or in the vicinity of aerodrome	Banks: At AD. Open H24. Post office: At AD. Open H24.
6	Tourist office	Tourist counter at AD.Office in the city: 154 Anna Salai, Chennai.Tel :28524295, 22854785
7	Remarks	Nil

VOMM AD 2.6 RESCUE AND FIRE FIGHTING SERVICES

1	Aerodrome category for fire fighting	Within ATS HR: CAT-9
2	Rescue equipment	AVBL as per category.
3	Capability for removal of disabled aircraft	1. The critical aircraft for the airport is B-744. 2. Primary responsibility for removal of disabled aircraft rests with the concerned airline. 3. The airlines may contact 'Aerodrome Coordinator for Disabled Aircraft Removal Operations' for coordinating with the agencies having Disabled Aircraft Recovery Kit. 4. Contact Details of Aerodrome Coordinator for Disabled Aircraft Removal Operations: Airport Director, Tel/Mob: 044-22561122 / 9940192926 Email: apdchennai@aai.aero
4	Remarks	NIL

VOMM AD 2.7 SEASONAL AVAILABILITY CLEARING

1	Type(s) of clearing equipment	Mechanical Sweeper
2	Clearance priorities	NIL
3	Remarks	NIL

VOMM AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA

1	Designation, surface and strength of aprons	Refer Aircraft Parking/Docking Chart
2	Designation, width, surface and strength of taxiways	Refer Aircraft Parking/Docking Chart
3	Location and elevation of altimeter checkpoints	Location: Domestic and International Apron Elevation 43 FT Location: TWY 'J' Elevation 51 FT

4	Location of VOR checkpoints	TWY F: 125908.07N 0800934.97E/ELEV 41 FT TWY J: 125940.00N 0801100.00E/ELEV 50 FT TWY K: 125902.09N 0800919.48E/ELEV42 FT TWY N: 130022.00N 0800920.92E/ELEV41 FT
5	Position of INS checkpoints	NIL

6	Remarks	**1. Coordinates of RWY Holding Position**

RWY Holding Position	Coordinates	Elevation (FT) (EGM08)
RHP I	125938.17N 0801044.47E	47
RHP II	125942.31N 0801055.49E	50
RHP III	125942.57N 0801044.45E	46
TWY A	125937.58N 0801054.15E	48
TWY B	125934.20N 0801050.82E	47
TWY C	125931.99N 0801036.49E	43
TWY D	125925.70N 0801019.78E	43
TWY F	125908.10N 0800934.85E	41
TWY J	125940.072N 0801100.07E	50
TWY K	125902.12N 0800919.38E	42
TWY L1	125954.28N 0801028.21E	36
TWY L3	125958.98N 0801019.13E	35
TWY M	125913.20N 0800947.57E	43
TWY N	130022.09N 0800920.97E	41
TWY N3	125939.31N 0801036.52E	42
TWY N5	125934.19N 0801042.22E	43
TWY P	125922.56N 0800952.13E	42
TWY P1	130004.13N 0800954.37E	44
TWY Q	125932.31N 0801018.89E	42
TWY Q1	125950.88N 0801019.83E	38
TWY T	125946.23N 0801044.23E	42
TWY T1	125944.56N 0801051.51E	48
TWY R3	125913.46N 0800930.10E	41
TWY S	125942.27N 0801037.78E	41
TWY U	125907.79N 0800932.31E	41

2. Coordinates of RWY & Taxi Intersection

TWY	Coordinates	Elevation(FT)
A	125940.02N 0801049.38E	51
C	125934.77N 0801035.40E	44
D & Q	125928.50N 0801018.69E	44
F	125909.56N 0800928.22E	43
J	125944.98N 0801102.59E	54
K	125903.84N 0800913.00E	43
L1	125951.64N 0801026.80E	38
L3	125956.31N 0801017.70E	38
M & P	125918.45N 0800951.92E	43
Q1	125955.12N 0801020.04E	38
T	125942.62N 0801044.35E	46
N	130024.72N 0800922.41E	41
N1	130018.71N 0800934.11E	43
N3 & N5	125936.92N 0801041.14E	45
P1	130008.19N 0800954.60E	43
R3	125943.84N 0801041.98E	45
S	125910.66N 0800931.19E	43
U	125910.67N 0800931.19E	43
V	125923.27N 0801004.82E	43
Y	125930.74N 0801024.65E	44
Z	125924.18N 0801007.21E	43

3. All aircraft stands are Power-in/Pushback. Due to non-availability of Power-in/Power-out aircraft stands, Non-Schedule operators should ensure availability of tow-bar on board or with Ground Handling Agents.
4. Non-Schedule operators requiring night parking to obtain prior clearance as per Night Parking Policy of AAI.

		5 . Simultaneous pushback from adjacent stands not permitted. Simultaneous push back is permitted to aircraft only from stands separated at least by one stand and when no aircraft is pushing back from any of the two rear stands (behind the pushing back aircraft depending upon the direction of push back). 6. Following multiple apron ramp system of parking is established at Parking Stand Nos. 27, 28, 29, 43, 44 and 45: i. Superimposed stands are designated as 27L, 27R, 28L, 28R, 29L, 29R, 43L, 43R, 44L, 44R and 45L. ii. Aircraft stands 27, 28, 29, 43, 44 and 45 are suitable for parking of up to CODE E type. iii. Lead in lines for the parking stands 27, 28, 29, 43, 44, and 45 are continuous yellow. iv. Secondary superimposed stands 27L, 27R, 28L, 28R, 29L, 29R, 43L, 43R, 44L, 44R and 45L are suitable for parking of up to CODE C type. v. Lead in lines for the secondary stands 27L, 27R, 28L, 28R, 29L, 29R, 43L, 43R, 44L, 44R and 45L are broken yellow. 7. Taxi lane H between TWY E and TWY D1 restricted up to CODE C.

VOMM AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

1	Use of aircraft stand identification signs, taxiway guidelines and visual docking/parking guidance system at aircraft stands	Taxiway guidelines from all intersections with taxiway and runway up to aircraft stands. Guidelines on Apron: Nose in guidance at parking stands with AVDGS (Camera type): Stand No. 27L, 27R, 28L, 28R, 29L, 29R, 30, 63, 64, 65, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 101, 102, 103 & 104 Guidelines on Apron: Nose in guidance at parking stands with SVDGS: Stand No. 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 35, 36, 43, 43L, 43R, 44, 44L, 45, 45L, 46, 47, 48, 49, 51, 52, 53, 54, 55, 56, 57
2	Runway and taxiway markings and lights	RWY Markings Designation, Centerline, THR, TDZ, Edge, Aiming Point, Side stripes, Transverse Stripe, Displaced THR on RWY 12 Lights THR, Centreline lights (RWY 07/25), Touch Down Zone (RWY 25), RETIL (TWY Z, TWY Y and TWY V), End and Edge. TWY Marking Centerline, Edge, RWY Holding Positions, Intermediate Holding Positions. Lights: Edge
3	Stop bars (if any)	NIL
4	Remarks	Runway guard lights have been provided at all RHP

VOMM AD 2.10 AERODROME OBSTACLES

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
25/TKOF 07/APCH	OTHER	125855.6N 0800906.1E	56 FT	NIL	Mobile Road Traffic (Road Elev.+ Traffic Height 12.2 + 5m)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
25/TKOF 07/APCH	TREE	125853.6N 0800901.8E	86 FT	NIL	TREE
25/TKOF 07/APCH	POLE	125903.2N 0800859.8E	63 FT	NIL	Electric Pole
25/TKOF 07/APCH	POLE	125904.1N 0800859.6E	63 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	TREE	125855.1N 0800854.5E	77 FT	NIL	TREE
25/TKOF 07/APCH	BUILDING	125903.0N 0800855.6E	81 FT	NIL	BUILDING
25/TKOF 07/APCH	BUILDING	125903.4N 0800855.9E	80 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	125903.0N 0800854.8E	75 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	125902.7N 0800855.1E	76 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	125903.2N 0800853.5E	78 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	125903.5N 0800853.6E	77 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	125903.4N 0800852.9E	85 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	125902.0N 0800850.1E	79 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	125859.1N 0800848.5E	83 FT	NIL	BUILDING
25/TKOF 07/APCH	TREE	125858.5N 0800848.5E	84 FT	NIL	Group of Trees
25/TKOF 07/APCH	TREE	125855.7N 0800846.3E	92 FT	NIL	Group of Trees
25/TKOF 07/APCH	TREE	125857.9N 0800847.3E	88 FT	NIL	Tree
25/TKOF 07/APCH	TREE	125858.8N 0800847.1E	91 FT	NIL	Group of Trees
25/TKOF 07/APCH	TREE	125859.3N 0800842.6E	99 FT	NIL	TREE
25/TKOF 07/APCH	TREE	125855.4N 0800845.0E	96 FT	NIL	Group of Trees
25/TKOF 07/APCH	TREE	125848.4N 0800841.7E	107 FT	NIL	Group of Trees
25/TKOF 07/APCH	BUILDING	125851.3N 0800845.0E	100 FT	NIL	BUILDING
25/TKOF 07/APCH	TREE	125900.2N 0800838.7E	104 FT	NIL	GROUP OF TREES
25/TKOF 07/APCH	TREE	125842.1N 0800838.0E	152 FT	NIL	GROUP OF TREES
25/TKOF 07/APCH	TREE	125848.3N 0800836.6E	115 FT	NIL	GROUP OF TREES
25/TKOF 07/APCH	OTHER	125808.5N 0800727.3E	226 FT	NIL	TREE ON HILL TOP
25/TKOF 07/APCH	OTHER	125807.0N 0800725.4E	262 FT	NIL	TREE ON HILL

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
25/TKOF 07/APCH	OTHER	125814.5N 0800743.0E	219 FT	NIL	CELL PHONE MAST
25/TKOF 07/APCH	OTHER	125900.3N 0800903.4E	55 FT	NIL	LIGHTNING ARRESTOR ON LLZ ANTENNA
25/TKOF 07/APCH	BUILDING	125904.2N 0800901.6E	69 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	125902.6N 0800854.9E	90 FT	NIL	LADDER ON BUILDING
25/TKOF 07/APCH	BUILDING	125859.1N 0800847.8E	84 FT	NIL	BUILDING
25/TKOF 07/APCH	BUILDING	125900.9N 0800850.4E	78 FT	NIL	BUILDING
25/TKOF 07/APCH	BUILDING	125901.4N 0800847.6E	86 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	125902.5N 0800851.6E	87 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	125902.3N 0800852.5E	80 FT	NIL	BUILDING U/C
25/TKOF 07/APCH	BUILDING	125903.3N 0800852.9E	85 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	125904.3N 0800900.6E	72 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	125903.9N 0800900.0E	63 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	125903.4N 0800900.0E	63 FT	NIL	BUILDING
25/TKOF 07/APCH	BUILDING	125848.1N 0800849.6E	94 FT	NIL	BUILDING
25/TKOF 07/APCH	BUILDING	125850.4N 0800844.8E	98 FT	NIL	SINTEX TANK ON BUILDING TOP
25/TKOF 07/APCH	TREE	125849.6N 0800849.1E	89 FT	NIL	SINTEX TANK ON BUILDING TOP
25/TKOF 07/APCH	TREE	125846.2N 0800848.0E	97 FT	NIL	Group of Trees
25/TKOF 07/APCH	TREE	125848.1N 0800844.2E	106 FT	NIL	Group of Trees
25/TKOF 07/APCH	TREE	125847.8N 0800848.4E	101 FT	NIL	GROUP OF TREES
25/TKOF 07/APCH	TREE	125851.9N 0800854.7E	78 FT	NIL	GROUP OF TREES
25/TKOF 07/APCH	BUILDING	125857.5N 0800845.9E	89 FT	NIL	SINTEX TANK ON BUILDING TOP
25/TKOF 07/APCH	TREE	125856.1N 0800846.1E	91 FT	NIL	Group of Trees
25/TKOF 07/APCH	TREE	125856.9N 0800845.9E	96 FT	NIL	Group of Trees
25/TKOF 07/APCH	TREE	125859.7N 0800842.5E	103 FT	NIL	GROUP OF TREES
25/TKOF 07/APCH	OTHER	125838.5N 0800828.5E	138 FT	NIL	MAST
25/TKOF 07/APCH	OTHER	125838.4N 0800814.0E	168 FT	NIL	Cell Phone Mast

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
25/TKOF 07/APCH	OTHER	125850.9N 0800809.3E	179 FT	NIL	CELL PHONE MAST
25/TKOF 07/APCH	BUILDING	125902.7N 0800851.0E	88 FT	NIL	TANK ON BUILDING (BUILDING TOP - 25.8 M)
25/TKOF 07/APCH	BUILDING	125901.4N 0800847.1E	84 FT	NIL	BUILDING
25/TKOF 07/APCH	OTHER	125858.0N 0800845.3E	93 FT	NIL	GROUP OF TREES
25/TKOF 07/APCH	ANTENNA	125901.7N 0800859.5E	64 FT	NIL	MLAT ANTENNA TOP
25/TKOF 07/APCH	OTHER	125905.6N 0800902.7E	67 FT	NIL	MOBILE ROAD TRAFFIC(ROAD ELEV . + TRAFFIC HEIGHT 15.3 + 5M)
25/TKOF 07/APCH	OTHER	125901.7N 0800859.4E	66 FT	NIL	LIGHTNING ARRESTOR ON DISH
25/APCH 07/TKOF	TREE	125953.6N 0801110.0E	94 FT	NIL	TREE
25/APCH 07/TKOF	POLE	125954.6N 0801114.3E	84 FT	NIL	LIGHT POST
25/APCH 07/TKOF	TREE	125954.9N 0801114.2E	122 FT	NIL	TREE
25/APCH 07/TKOF	TREE	125956.8N 0801119.8E	118 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	125956.2N 0801121.0E	90 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	125955.3N 0801120.2E	103 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	125953.5N 0801121.6E	102 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	TREE	125954.2N 0801122.6E	110 FT	NIL	Group of Tree
25/APCH 07/TKOF	TREE	125956.6N 0801122.8E	97 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	TREE	125957.4N 0801123.4E	109 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	TREE	125958.3N 0801124.4E	114 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	TREE	125955.4N 0801123.8E	101 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	125954.2N 0801124.7E	105 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	125953.6N 0801125.7E	112 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	125953.1N 0801124.2E	105 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	125952.6N 0801123.1E	97 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	125952.2N 0801125.7E	122 FT	NIL	Group of Trees

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
25/APCH 07/TKOF	TREE	125950.7N 0801122.3E	95 FT	NIL	GROUP OF TREES
25/APCH 07/TKOF	OTHER	125947.3N 0801122.2E	90 FT	NIL	MOBILE METRO TRAFFIC (BRIDGE ELEV .+ TRAFFIC HEIGHT 19.7 + 7.6M)
25/APCH 07/TKOF	SIGN	125947.6N 0801125.3E	96 FT	NIL	HOARDING
25/APCH 07/TKOF	TREE	125954.9N 0801130.4E	107 FT	NIL	TREE
25/APCH 07/TKOF	TREE	125954.1N 0801128.0E	134 FT	NIL	TREE
25/APCH 07/TKOF	TREE	125955.5N 0801126.9E	120 FT	NIL	TREE
25/APCH 07/TKOF	TREE	125956.0N 0801128.0E	115 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	125958.6N 0801128.1E	117 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	125958.8N 0801129.0E	116 FT	NIL	Group of Trees
25/APCH 07/TKOF	OTHER	130000.9N 0801127.5E	106 FT	NIL	FLOOD LIGHT MAST
25/APCH 07/TKOF	TREE	130001.4N 0801126.5E	109 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	130002.3N 0801127.6E	114 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	130001.5N 0801127.8E	106 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	130000.4N 0801129.7E	120 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	TREE	130000.0N 0801129.9E	117 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	TREE	125957.0N 0801130.1E	110 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	125957.6N 0801131.0E	125 FT	NIL	TREE
25/APCH 07/TKOF	TREE	125958.1N 0801131.4E	120 FT	NIL	Group of Trees
25/APCH 07/TKOF	OTHER	125957.6N 0801136.2E	133 FT	NIL	MOBILE METRO TRAFFIC (BRIDGE ELEV. + TRAFFIC HEIGHT 32.8 + 7.6M)
25/APCH 07/TKOF	TREE	130000.2N 0801136.6E	122 FT	NIL	TREE
25/APCH 07/TKOF	TREE	125959.1N 0801136.3E	134 FT	NIL	TREE
25/APCH 07/TKOF	TREE	125957.1N 0801133.3E	130 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	125957.7N 0801133.1E	125 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	TREE	130000.0N 0801132.0E	147 FT	NIL	GROUP OF TREE

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
25/APCH 07/TKOF	TREE	130000.0N 0801135.6E	135 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	TREE	130000.9N 0801135.6E	128 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	TREE	130000.6N 0801135.2E	144 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	BUILDING	130001.3N 0801135.1E	147 FT	NIL	BUILDING
25/APCH 07/TKOF	TREE	130010.7N 0801142.4E	139 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	OTHER	130003.1N 0801137.6E	130 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	OTHER	130003.2N 0801139.8E	134 FT	NIL	GROUP OF TREES
25/APCH 07/TKOF	TREE	130008.2N 0801136.2E	149 FT	NIL	OVER HEAD WATER TANK
25/APCH 07/TKOF	OTHER	130000.5N 0801124.7E	129 FT	NIL	Cell Phone Mast
25/APCH 07/TKOF	BUILDING	130002.9N 0801127.6E	111 FT	NIL	BUILDING
25/APCH 07/TKOF	TREE	125958.5N 0801122.6E	144 FT	NIL	GROUP OF TREES
25/APCH 07/TKOF	TREE	130002.6N 0801129.6E	135 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	TREE	130001.4N 0801131.3E	143 FT	NIL	GROUP OF TREES
25/APCH 07/TKOF	TREE	125959.4N 0801132.4E	143 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	TREE	130001.3N 0801135.1E	130 FT	NIL	GROUP OF TREES
25/APCH 07/TKOF	TREE	125958.9N 0801129.7E	117 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	TREE	130008.4N 0801138.2E	150 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	TANK	130006.6N 0801133.6E	129 FT	NIL	OVER HEAD WATER TANK
25/APCH 07/TKOF	FENCE	125951.8N 0801106.7E	66 FT	NIL	AIRPORT BOUNDARY WALL WITH FENCING ON TOP
25/APCH 07/TKOF	TREE	125954.7N 0801113.5E	116 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	TREE	125953.9N 0801111.5E	96 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	125955.8N 0801115.3E	130 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	125955.4N 0801117.3E	90 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	125957.5N 0801121.2E	149 FT	NIL	Building (Base Elev.-12.4m)
25/APCH 07/TKOF	TREE	125951.8N 0801120.7E	89 FT	NIL	GROUP OF TREES

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
25/APCH 07/TKOF	TREE	125951.4N 0801122.9E	99 FT	NIL	Group of Trees
25/APCH 07/TKOF	TREE	125950.3N 0801124.4E	94 FT	NIL	GROUP OF TREE
25/APCH 07/TKOF	POLE	125945.7N 0801119.9E	80 FT	NIL	METRO ELECTRIC LINE POLE
25/APCH 07/TKOF	POLE	125953.9N 0801112.5E	76 FT	NIL	LIGHT POST
25/APCH 07/TKOF	POLE	125953.3N 0801110.9E	77 FT	NIL	LIGHT POST
25/APCH 07/TKOF	POLE	125952.7N 0801109.4E	80 FT	NIL	LIGHT POST
25/APCH 07/TKOF	POLE	125952.1N 0801107.8E	77 FT	NIL	LIGHT POST
25/APCH 07/TKOF	OTHER	125951.5N 0801106.7E	72 FT	NIL	MOBILE ROAD TRAFFIC (ROAD ELEV. 16.9 M + 5 M TRAFFIC HT.)
25/APCH 07/TKOF	OTHER	125943.1N 0801112.6E	65 FT	NIL	MOBILE ROAD TRAFFIC (ROAD ELEV . + TRAFFIC HEIGHT 14.7 + 5M)
25/APCH 07/TKOF	POLE	125952.0N 0801107.0E	76 FT	NIL	LIGHT POLE
25/APCH 07/TKOF	POLE	125952.4N 0801108.6E	78 FT	NIL	LIGHT POLE
25/APCH 07/TKOF	POLE	125953.0N 0801110.1E	78 FT	NIL	LIGHT POLE
25/APCH 07/TKOF	TREE	125959.8N 0801125.7E	130 FT	NIL	GROUP OF TREES
25/APCH 07/TKOF	BUILDING	130003.1N 0801128.0E	112 FT	NIL	BUILDING
30/TKOF 12/APCH	OTHER	130027.7N 0800922.1E	56 FT	NIL	MOBILE ROAD TRAFFIC (ROAD ELEV. 12.1M + 5.0M TRAFFIC HEIGHT)
30/TKOF 12/APCH	BUILDING	130028.3N 0800921.2E	67 FT	NIL	SHED TOP
30/TKOF 12/APCH	TREE	130027.2N 0800911.6E	76 FT	NIL	TREE
30/TKOF 12/APCH	BUILDING	130032.2N 0800917.5E	77 FT	NIL	BUILDING DOME TOP
30/TKOF 12/APCH	BUILDING	130032.8N 0800918.0E	70 FT	NIL	BUILDING
30/TKOF 12/APCH	BUILDING	130032.2N 0800917.2E	76 FT	NIL	WATER TANK ON BUILDING TOP
30/TKOF 12/APCH	BUILDING	130032.6N 0800916.8E	73 FT	NIL	WATER TANK ON BUILDING TOP
30/TKOF 12/APCH	BUILDING	130030.6N 0800914.1E	73 FT	NIL	WATER TANK ON BUILDING
30/TKOF 12/APCH	BUILDING	130030.2N 0800913.6E	73 FT	NIL	WATER TANK ON BUILDING

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
30/TKOF 12/APCH	TREE	130029.8N 0800912.0E	86 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TREE	130030.6N 0800911.5E	84 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TREE	130031.7N 0800912.9E	91 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TREE	130029.2N 0800904.5E	90 FT	NIL	TREE
30/TKOF 12/APCH	TREE	130027.4N 0800906.8E	105 FT	NIL	TREE
30/TKOF 12/APCH	TREE	130033.8N 0800904.8E	93 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	BRIDGE	130037.0N 0800904.3E	95 FT	NIL	WATER TANK ON BUILDING TOP
30/TKOF 12/APCH	BUILDING	130038.1N 0800905.8E	97 FT	NIL	TANK ON BUILDING (BUILDING TOP - 28.4 M)
30/TKOF 12/APCH	POLE	130038.6N 0800907.0E	104 FT	NIL	POLE
30/TKOF 12/APCH	BUILDING	130038.6N 0800908.4E	90 FT	NIL	BUILDING
30/TKOF 12/APCH	BUILDING	130040.2N 0800909.8E	98 FT	NIL	BUILDING
30/TKOF 12/APCH	TREE	130030.6N 0800915.1E	82 FT	NIL	Group of Trees
30/TKOF 12/APCH	OTHER	130035.5N 0800912.2E	101 FT	NIL	CELLPHONE MAST ON BUILDING
30/TKOF 12/APCH	BUILDING	130029.9N 0800921.3E	63 FT	NIL	BUILDING WHITE SHED TOP
30/TKOF 12/APCH	BUILDING	130035.8N 0800912.9E	80 FT	NIL	Building
30/TKOF 12/APCH	BUILDING	130033.0N 0800912.6E	79 FT	NIL	BUILDING
30/TKOF 12/APCH	BUILDING	130035.6N 0800916.1E	87 FT	NIL	BUILDING
30/TKOF 12/APCH	TREE	130033.0N 0800916.3E	81 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TREE	130030.7N 0800916.3E	97 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	BUILDING	130035.5N 0800914.3E	96 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TREE	130034.5N 0800907.9E	108 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TREE	130041.7N 0800903.3E	119 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	FENCE	130029.1N 0800925.2E	52 FT	NIL	AIRPORT BOUNDARY WALL WITH FENCING ON TOP
30/TKOF 12/APCH	POLE	130027.1N 0800915.5E	67 FT	NIL	ELECTRIC POLE

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
30/TKOF 12/APCH	POLE	130028.5N 0800915.9E	67 FT	NIL	ELECTRIC POLE
30/TKOF 12/APCH	BUILDING	130028.6N 0800919.6E	61 FT	NIL	SHED TOP
30/TKOF 12/APCH	TREE	130031.0N 0800916.7E	93 FT	NIL	Group of Tree
30/TKOF 12/APCH	TREE	130032.3N 0800916.0E	95 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TREE	130030.8N 0800912.8E	97 FT	NIL	TREE
30/TKOF 12/APCH	BUILDING	130029.3N 0800921.4E	63 FT	NIL	BUILDING
30/TKOF 12/APCH	BUILDING	130029.3N 0800917.6E	80 FT	NIL	BUILDING
30/TKOF 12/APCH	BUILDING	130032.6N 0800915.7E	75 FT	NIL	BUILDING
30/TKOF 12/APCH	BUILDING	130034.0N 0800911.8E	85 FT	NIL	BUILDING
30/TKOF 12/APCH	BUILDING	130036.5N 0800903.8E	101 FT	NIL	BUILDING
30/TKOF 12/APCH	BRIDGE	130036.5N 0800903.8E	101 FT	NIL	BUILDING
30/TKOF 12/APCH	BUILDING	130039.0N 0800907.6E	100 FT	NIL	BUILDING
30/TKOF 12/APCH	BUILDING	130035.5N 0800902.4E	98 FT	NIL	BUILDING
30/TKOF 12/APCH	BRIDGE	130039.9N 0800901.9E	102 FT	NIL	BUILDING U/C
30/TKOF 12/APCH	BUILDING	130029.4N 0800906.2E	86 FT	NIL	BUILDING
30/TKOF 12/APCH	BUILDING	130028.2N 0800910.5E	81 FT	NIL	BUILDING U/C
30/TKOF 12/APCH	BUILDING	130027.0N 0800912.1E	78 FT	NIL	BUILDING U/C
30/TKOF 12/APCH	BUILDING	130033.6N 0800902.4E	96 FT	NIL	BUILDING
30/TKOF 12/APCH	OTHER	130039.1N 0800905.3E	110 FT	NIL	FLOOD LIGHT MAST
30/TKOF 12/APCH	OTHER	130036.1N 0800857.4E	107 FT	NIL	FLOOD LIGHT MAST
30/TKOF 12/APCH	TREE	130033.2N 0800907.4E	92 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TREE	130033.9N 0800910.3E	102 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TREE	130036.4N 0800906.5E	106 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TREE	130034.6N 0800902.0E	99 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TREE	130043.2N 0800903.0E	124 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TREE	130030.6N 0800902.0E	103 FT	NIL	GROUP OF TREES

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
30/TKOF 12/APCH	TREE	130031.1N 0800902.6E	110 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TREE	130031.5N 0800903.0E	108 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TANK	130049.2N 0800857.0E	125 FT	NIL	OVER HEAD WATER TANK
30/TKOF 12/APCH	TREE	130033.4N 0800910.4E	89 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	BUILDING	130030.2N 0800913.8E	75 FT	NIL	BUILDING
30/TKOF 12/APCH	OTHER	130028.2N 0800923.7E	57 FT	NIL	MOBILE ROAD TRAFFIC (ROAD ELEV. 12.4 M + 5 M TRAFFIC HT.)
30/TKOF 12/APCH	BUILDING	130032.0N 0800918.0E	80 FT	NIL	BUILDING
30/TKOF 12/APCH	BUILDING	130034.1N 0800917.5E	80 FT	NIL	BUILDING
30/TKOF 12/APCH	BRIDGE	130031.2N 0800912.5E	78 FT	NIL	TANK ON BUILDING (BUILDING TOP - 22.5 M)
30/TKOF 12/APCH	BUILDING	130029.0N 0800907.3E	93 FT	NIL	BUILDING
30/TKOF 12/APCH	BUILDING	130031.7N 0800906.2E	87 FT	NIL	GRILL ON BUILDING (TANK TOP - 25.7 M)
30/TKOF 12/APCH	BUILDING	130030.6N 0800905.9E	87 FT	NIL	BUILDING (U/C)
30/TKOF 12/APCH	BUILDING	130031.0N 0800905.4E	92 FT	NIL	BUILDING
30/TKOF 12/APCH	BUILDING	130032.0N 0800903.4E	94 FT	NIL	Building
30/TKOF 12/APCH	TREE	130028.0N 0800921.5E	65 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TREE	113029.0N 0800921.6E	70 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TREE	130028.7N 0800920.0E	66 FT	NIL	TREE
30/TKOF 12/APCH	BUILDING	130032.2N 0800918.9E	68 FT	NIL	Building
30/TKOF 12/APCH	TREE	130032.5N 0800917.5E	97 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	BUILDING	130032.1N 0800916.6E	79 FT	NIL	BUILDING
30/TKOF 12/APCH	TREE	130030.7N 0800915.5E	91 FT	NIL	Group of Trees
30/TKOF 12/APCH	OTHER	130031.8N 0800916.0E	78 FT	NIL	SOLAR LIGHT
30/TKOF 12/APCH	TREE	130030.6N 0800914.3E	86 FT	NIL	Group of Trees
30/TKOF 12/APCH	TREE	130034.3N 0800915.5E	94 FT	NIL	Group of Trees

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
30/TKOF 12/APCH	BUILDING	130036.0N 0800915.2E	94 FT	NIL	BUILDING
30/TKOF 12/APCH	TREE	130034.5N 0800914.0E	90 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	BUILDING	130028.0N 0800905.7E	85 FT	NIL	SINTEX TANK ON BUILDING
30/TKOF 12/APCH	BUILDING	130036.7N 0800914.3E	94 FT	NIL	WATER TANK ON BUILDING
30/TKOF 12/APCH	TREE	130033.9N 0800912.8E	84 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	POLE	130030.9N 0800911.0E	78 FT	NIL	ELECTRIC POLE
30/TKOF 12/APCH	BUILDING	130034.8N 0800911.8E	80 FT	NIL	BLUE SHED TOP
30/TKOF 12/APCH	TREE	130026.6N 0800905.6E	94 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	BUILDING	130038.7N 0800912.0E	92 FT	NIL	SINTEX TANK ON BUILDING
30/TKOF 12/APCH	TREE	130035.3N 0800909.7E	91 FT	NIL	Group of Trees
30/TKOF 12/APCH	TREE	130025.8N 0800903.3E	100 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TREE	130035.2N 0800908.3E	92 FT	NIL	Tree
30/TKOF 12/APCH	TREE	130028.0N 0800903.6E	101 FT	NIL	TREE
30/TKOF 12/APCH	TREE	130037.2N 0800907.7E	95 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	TREE	130035.0N 0800905.2E	94 FT	NIL	TREE
30/TKOF 12/APCH	BUILDING	130039.8N 0800906.0E	96 FT	NIL	Building
30/TKOF 12/APCH	BUILDING	130035.7N 0800902.9E	100 FT	NIL	Building
30/TKOF 12/APCH	TREE	130036.4N 0800902.3E	105 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	BUILDING	130034.6N 0800900.7E	103 FT	NIL	SINTEX TANK ON BUILDING
30/TKOF 12/APCH	TREE	130043.3N 0800900.0E	117 FT	NIL	GROUP OF TREES
30/TKOF 12/APCH	BUILDING	130050.3N 0800855.9E	129 FT	NIL	BUILDING U/C
30/APCH 12/TKOF	TREE	125931.1N 0801058.5E	94 FT	NIL	GROUP OF TREES
30/APCH 12/TKOF	TREE	125936.8N 0801108.7E	92 FT	NIL	TREE
30/APCH 12/TKOF	SIGN	125937.7N 0801107.4E	85 FT	NIL	BOARD
30/APCH 12/TKOF	OTHER	125937.5N 0801110.1E	92 FT	NIL	LIGHTNING ARRESTOR ON BUILDING (BUILDING TOP - 26.6 M)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
30/APCH 12/TKOF	TREE	125936.4N 0801105.2E	72 FT	NIL	TREE
30/APCH 12/TKOF	POLE	125937.2N 0801104.7E	74 FT	NIL	ELECTRIC POLE
30/APCH 12/TKOF	OTHER	125935.0N 0801059.2E	62 FT	NIL	MOBILE ROAD TRAFFIC (ROAD ELEV.+ TRAFFIC HEIGHT 13.9 + 5M)
30/APCH 12/TKOF	POLE	125931.7N 0801056.3E	59 FT	NIL	LIGHT POST
30/APCH 12/TKOF	TREE	125931.9N 0801057.4E	88 FT	NIL	TREE
30/APCH 12/TKOF	TREE	125932.3N 0801058.8E	84 FT	NIL	TREE
30/APCH 12/TKOF	POLE	125932.2N 0801058.2E	73 FT	NIL	ELECTRIC POLE
30/APCH 12/TKOF	TREE	125931.4N 0801059.4E	81 FT	NIL	TREE
30/APCH 12/TKOF	BUILDING	125932.6N 0801102.1E	65 FT	NIL	WATER TANK ON BUILDING TOP
30/APCH 12/TKOF	BUILDING	125927.0N 0801102.0E	94 FT	NIL	BUILDING
30/APCH 12/TKOF	BUILDING	125927.4N 0801102.0E	94 FT	NIL	WATER TANK ON BUILDING TOP
30/APCH 12/TKOF	TREE	125925.9N 0801103.4E	98 FT	NIL	Group of Trees
30/APCH 12/TKOF	BUILDING	125926.6N 0801104.2E	84 FT	NIL	BUILDING
30/APCH 12/TKOF	TREE	125928.4N 0801106.6E	90 FT	NIL	TREE
30/APCH 12/TKOF	TREE	125929.0N 0801106.8E	90 FT	NIL	Group of Trees
30/APCH 12/TKOF	TREE	130041.7N 0800903.3E	73 FT	NIL	TREE
30/APCH 12/TKOF	TREE	125931.4N 0801107.2E	78 FT	NIL	TREE
30/APCH 12/TKOF	TREE	125931.5N 0801107.8E	88 FT	NIL	TREE
30/APCH 12/TKOF	TREE	125932.1N 0801107.7E	76 FT	NIL	TREE
30/APCH 12/TKOF	TREE	125932.8N 0801107.3E	82 FT	NIL	TREE
30/APCH 12/TKOF	TREE	125932.6N 0801110.1E	84 FT	NIL	GROUP OF TREES
30/APCH 12/TKOF	TANK	125932.7N 0801121.3E	115 FT	NIL	OVER HEAD WATER TANK
30/APCH 12/TKOF	TREE	125933.7N 0801115.6E	100 FT	NIL	TREE
30/APCH 12/TKOF	TREE	125933.2N 0801112.0E	100 FT	NIL	GROUP OF TREES
30/APCH 12/TKOF	TREE	125926.9N 0801106.3E	79 FT	NIL	TREE

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
30/APCH 12/TKOF	TREE	125924.6N 0801107.0E	103 FT	NIL	TREE
30/APCH 12/TKOF	BUILDING	125924.9N 0801104.4E	96 FT	NIL	Building
30/APCH 12/TKOF	OTHER	125938.7N 0801101.0E	72 FT	NIL	WIND DIRECTION INDICATOR
30/APCH 12/TKOF	POLE	125937.3N 0801107.2E	73 FT	NIL	LIGHT POST
30/APCH 12/TKOF	POLE	125934.4N 0801101.8E	65 FT	NIL	ELECTRIC POLE
30/APCH 12/TKOF	POLE	125933.5N 0801100.1E	64 FT	NIL	ELECTRIC POLE
30/APCH 12/TKOF	POLE	125932.9N 0801059.2E	64 FT	NIL	ELECTRIC POLE
30/APCH 12/TKOF	POLE	125931.3N 0801055.8E	81 FT	NIL	LIGHT POST
30/APCH 12/TKOF	POLE	125932.0N 0801056.8E	59 FT	NIL	LIGHT POST
30/APCH 12/TKOF	POLE	125933.7N 0801059.3E	59 FT	NIL	LIGHT POST
30/APCH 12/TKOF	POLE	125933.3N 0801058.7E	63 FT	NIL	CAMERA POLE
30/APCH 12/TKOF	POLE	125935.0N 0801101.6E	62 FT	NIL	CAMERA POLE
30/APCH 12/TKOF	TREE	125937.4N 0801107.5E	82 FT	NIL	GROUP OF TREES
30/APCH 12/TKOF	TREE	125935.6N 0801103.9E	73 FT	NIL	GROUP OF TREES
30/APCH 12/TKOF	TREE	125934.8N 0801102.5E	71 FT	NIL	GROUP OF TREES
30/APCH 12/TKOF	TREE	125933.9N 0801102.9E	74 FT	NIL	GROUP OF TREES
30/APCH 12/TKOF	TREE	125933.9N 0801100.8E	88 FT	NIL	GROUP OF TREES
30/APCH 12/TKOF	TREE	125933.5N 0801100.1E	81 FT	NIL	GROUP OF TREES
30/APCH 12/TKOF	TREE	125932.2N 0801058.1E	89 FT	NIL	GROUP OF TREES
30/APCH 12/TKOF	OTHER	125934.5N 0801100.2E	64 FT	NIL	MOBILE ROAD TRAFFIC (ROAD ELEV.+ TRAFFIC HEIGHT 14.6 + 5M)
30/APCH 12/TKOF	FENCE	125933.6N 0801057.2E	57 FT	NIL	AIRPORT BOUNDARY WALL WITH FENCING ON TOP
30/APCH 12/TKOF	OTHER	125934.9N 0801100.8E	64 FT	NIL	MOBILE ROAD TRAFFIC (ROAD ELEV.+ TRAFFIC HEIGHT 14.5 + 5M)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
30/APCH 12/TKOF	OTHER	125932.2N 0801056.3E	64 FT	NIL	MOBILE ROAD TRAFFIC (ROAD ELEV.+ TRAFFIC HEIGHT 14.5 + 5M)
30/APCH 12/TKOF	BUILDING	125936.0N 0801112.0E	92 FT	NIL	BUILDING
30/APCH 12/TKOF	OTHER	125923.8N 0801136.7E	136 FT	NIL	CELL PHONE MAST
30/APCH 12/TKOF	BUILDING	125933.2N 0801112.9E	89 FT	NIL	BUILDING
30/APCH 12/TKOF	BUILDING	125932.8N 0801114.3E	87 FT	NIL	BUILDING
30/APCH 12/TKOF	TREE	125922.3N 0801111.5E	98 FT	NIL	GROUP OF TREES
30/APCH 12/TKOF	TREE	125933.0N 0801113.7E	97 FT	NIL	GROUP OF TREES
30/APCH 12/TKOF	TREE	125932.6N 0801112.2E	99 FT	NIL	GROUP OF TREES
30/APCH 12/TKOF	TREE	125931.2N 0801113.1E	92 FT	NIL	Group of Trees
30/APCH 12/TKOF	TREE	125930.6N 0801108.3E	94 FT	NIL	Group of Trees
30/APCH 12/TKOF	TREE	125925.7N 0801108.4E	108 FT	NIL	Group of Trees
30/APCH 12/TKOF	OTHER	125934.6N 0801114.8E	115 FT	NIL	CELL PHONE MAST
30/APCH 12/TKOF	BUILDING	125930.1N 0801058.0E	91 FT	NIL	BUILDING
30/APCH 12/TKOF	BUILDING	125928.5N 0801100.2E	92 FT	NIL	BUILDING
30/APCH 12/TKOF	BUILDING	125921.6N 0801110.3E	96 FT	NIL	BUILDING
30/APCH 12/TKOF	BUILDING	125920.1N 0801111.4E	97 FT	NIL	BUILDING
30/APCH 12/TKOF	TREE	125930.1N 0801058.2E	100 FT	NIL	Group of Trees
30/APCH 12/TKOF	OTHER	125936.8N 0801106.0E	72 FT	NIL	IOCL PETROL PUMP
30/APCH 12/TKOF	TREE	125934.9N 0801106.2E	88 FT	NIL	Group of Trees
30/APCH 12/TKOF	TREE	125933.4N 0801104.1E	88 FT	NIL	Group of Trees
30/APCH 12/TKOF	TREE	125935.5N 0801108.0E	100 FT	NIL	Group of Trees
30/APCH 12/TKOF	TREE	125930.6N 0801100.4E	88 FT	NIL	Group of Trees
30/APCH 12/TKOF	TREE	125927.8N 0801101.4E	104 FT	NIL	Group of Trees
30/APCH 12/TKOF	TREE	125930.2N 0801106.9E	89 FT	NIL	Group of Trees

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
30/APCH 12/TKOF	FENCE	125932.2N 0801055.5E	56 FT	NIL	AIRPORT BOUNDARY WALL WITH FENCING ON TOP
30/APCH 12/TKOF	FENCE	125932.9N 0801053.5E	58 FT	NIL	AIRPORT BOUNDARY WALL WITH FENCING ON TOP
30/APCH 12/TKOF	FENCE	125932.8N 0801053.4E	53 FT	NIL	AIRPORT BOUNDARY WALL WITH FENCING ON TOP
30/APCH 12/TKOF	TREE	125931.7N 0801057.1E	92 FT	NIL	Group of Trees
30/APCH 12/TKOF	TREE	125926.2N 0801104.5E	98 FT	NIL	Group of Trees
30/APCH 12/TKOF	TREE	125933.8N 0801107.8E	89 FT	NIL	Group of Trees
30/APCH 12/TKOF	TREE	125935.5N 0801109.3E	99 FT	NIL	Group of Trees
30/APCH 12/TKOF	TREE	125935.9N 0801111.4E	94 FT	NIL	Group of Trees
30/APCH 12/TKOF	OTHER	125934.8N 0801115.0E	109 FT	NIL	MAST
In circling area and at AD	OTHER	125902.6N 0800909.7E	44 FT	NIL	APPROACH LIGHT
In circling area and at AD	OTHER	125902.3N 0800908.8E	44 FT	NIL	APPROACH LIGHT
In circling area and at AD	TREE	125857.8N 0800921.0E	105 FT	NIL	GROUP OF TREES
In circling area and at AD	OTHER	125924.7N 0800829.6E	222 FT	NIL	CELL PHONE MAST
In circling area and at AD	TREE	125854.8N 0800905.9E	73 FT	NIL	Group of Trees
In circling area and at AD	TREE	125854.4N 0800904.8E	82 FT	NIL	TREE
In circling area and at AD	BUILDING	125904.7N 0800856.1E	74 FT	NIL	VENTILATION PIPE ON BUILDING
In circling area and at AD	BUILDING	125904.6N 0800852.9E	90 FT	NIL	BUILDING
In circling area and at AD	BUILDING	125903.1N 0800850.3E	86 FT	NIL	BUILDING
In circling area and at AD	BUILDING	125802.7N 0800749.8E	204 FT	NIL	BUILDING
In circling area and at AD	BUILDING	125803.5N 0800747.9E	203 FT	NIL	BUILDING
In circling area and at AD	BUILDING	125807.0N 0800748.7E	208 FT	NIL	BUILDING
In circling area and at AD	TREE	125858.4N 0800919.1E	93 FT	NIL	GROUP OF TREES
In circling area and at AD	TREE	125858.4N 0800917.1E	94 FT	NIL	GROUP OF TREES
In circling area and at AD	TREE	125855.2N 0800906.6E	89 FT	NIL	GROUP OF TREES

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	125904.4N 0800853.8E	87 FT	NIL	BUILDING (BASE ELEV.-12.4M)
In circling area and at AD	BUILDING	125903.9N 0800854.8E	74 FT	NIL	BUILDING
In circling area and at AD	TREE	125908.3N 0800905.7E	75 FT	NIL	GROUP OF TREES
In circling area and at AD	TREE	125845.6N 0800848.7E	103 FT	NIL	GROUP OF TREES
In circling area and at AD	OTHER	125911.5N 0800923.4E	49 FT	NIL	MET INSTRUMENT
In circling area and at AD	OTHER	125912.0N 0800924.2E	49 FT	NIL	MET INSTRUMENT
In circling area and at AD	OTHER	125911.2N 0800922.3E	50 FT	NIL	MET INSTRUMENT
In circling area and at AD	ANTENNA	125911.1N 0800921.9E	76 FT	NIL	CIWS ANTENNA
In circling area and at AD	OTHER	125911.5N 0800921.8E	62 FT	NIL	LIGHTNING ARRESTOR
In circling area and at AD	BUILDING	125910.8N 0800920.3E	53 FT	NIL	GP HUT
In circling area and at AD	NAVAID	125910.9N 0800920.0E	57 FT	NIL	GP DME ANTENNA
In circling area and at AD	ANTENNA	125911.3N 0800921.8E	60 FT	NIL	MET ANTENNA
In circling area and at AD	TREE	125856.4N 0800915.9E	118 FT	NIL	GROUP OF TREES
In circling area and at AD	TREE	125852.7N 0800906.6E	106 FT	NIL	GROUP OF TREES
In circling area and at AD	BUILDING	125904.6N 0800857.7E	75 FT	NIL	Building
In circling area and at AD	BUILDING	125904.4N 0800852.8E	88 FT	NIL	BUILDING
In circling area and at AD	BUILDING	125902.8N 0800849.6E	84 FT	NIL	BUILDING
In circling area and at AD	ANTENNA	125911.0N 0800921.7E	74 FT	NIL	DCWIS ANTENNA TOP
In circling area and at AD	TREE	125951.5N 0801103.5E	120 FT	NIL	TREE
In circling area and at AD	TREE	125952.8N 0801105.3E	110 FT	NIL	TREE
In circling area and at AD	POLE	125952.0N 0801105.8E	103 FT	NIL	LIGHT POST
In circling area and at AD	TREE	125952.1N 0801106.4E	117 FT	NIL	TREE
In circling area and at AD	TREE	130003.5N 0801126.6E	131 FT	NIL	GROUP OF TREE
In circling area and at AD	NATURAL HIGHPOINT	130006.7N 0801127.9E	152 FT	NIL	HILL TOP
In circling area and at AD	TREE	130018.7N 0801137.8E	304 FT	NIL	Group of Trees
In circling area and at AD	BUILDING	130019.2N 0801138.2E	300 FT	NIL	CHURCH STOOB

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	ANTENNA	130019.1N 0801137.2E	301 FT	NIL	ANTENNA
In circling area and at AD	BUILDING	130017.5N 0801137.8E	287 FT	NIL	CHURCH LIGHTNING ARRESTER TOP
In circling area and at AD	TREE	130017.9N 0801137.3E	290 FT	NIL	GROUP OF TREE
In circling area and at AD	OTHER	130002.7N 0801124.5E	147 FT	NIL	CELL PHONE MAST
In circling area and at AD	BUILDING	125946.8N 0801140.5E	151 FT	NIL	Building
In circling area and at AD	BUILDING	130003.8N 0801127.7E	118 FT	NIL	Building
In circling area and at AD	BUILDING	130010.7N 0801132.6E	192 FT	NIL	SHED
In circling area and at AD	BUILDING	130017.2N 0801137.3E	276 FT	NIL	Building
In circling area and at AD	BUILDING	130018.0N 0801136.0E	292 FT	NIL	CHURCH
In circling area and at AD	TREE	130005.2N 0801127.9E	156 FT	NIL	GROUP OF TREE
In circling area and at AD	TREE	130014.3N 0801137.4E	204 FT	NIL	GROUP OF TREE
In circling area and at AD	TREE	130011.6N 0801134.2E	201 FT	NIL	Group of Trees
In circling area and at AD	TREE	130010.5N 0801132.4E	204 FT	NIL	Group of Trees
In circling area and at AD	TREE	130009.7N 0801129.5E	196 FT	NIL	Group of Trees
In circling area and at AD	TREE	130006.3N 0801128.6E	164 FT	NIL	Group of Trees
In circling area and at AD	NATURAL HIGHPOINT	130017.7N 0801136.7E	295 FT	NIL	HILL TOP
In circling area and at AD	OTHER	130016.3N 0801137.3E	270 FT	NIL	LIGHTNING ARRESTOR
In circling area and at AD	TREE	125952.1N 0801104.4E	118 FT	NIL	Group of Trees
In circling area and at AD	TREE	125952.7N 0801107.8E	106 FT	NIL	Group of Trees
In circling area and at AD	POLE	125955.0N 0801109.9E	106 FT	NIL	GROUP OF TREE
In circling area and at AD	BRIDGE	125945.0N 0801123.6E	97 FT	NIL	BUILDING
In circling area and at AD	BUILDING	125945.3N 0801124.5E	94 FT	NIL	Building
In circling area and at AD	OTHER	125944.6N 0801123.3E	118 FT	NIL	CELL PHONE MAST
In circling area and at AD	OTHER	125945.1N 0801125.1E	116 FT	NIL	CELL PHONE MAST
In circling area and at AD	SIGN	125944.4N 0801121.2E	89 FT	NIL	HOARDING
In circling area and at AD	OTHER	130004.0N 0801126.2E	154 FT	NIL	CELL PHONE MAST

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	130010.1N 0801129.0E	193 FT	NIL	SCHOOL
In circling area and at AD	POLE	125951.9N 0801106.2E	79 FT	NIL	LIGHT POST
In circling area and at AD	ANTENNA	125945.1N 0801053.6E	64 FT	NIL	ANTENNA ON GP BUILDING (BUILDING TOP - 18.0 M)
In circling area and at AD	OTHER	125939.4N 0801102.7E	81 FT	NIL	LIGHTNING ARRESTOR
In circling area and at AD	ANTENNA	125940.0N 0801103.0E	84 FT	NIL	CIWS ANTENNA
In circling area and at AD	TREE	125940.8N 0801109.6E	92 FT	NIL	GROUP OF TREE
In circling area and at AD	TREE	125940.2N 0801107.7E	88 FT	NIL	Tree
In circling area and at AD	ANTENNA	125945.3N 0801053.8E	83 FT	NIL	ANTENNA
In circling area and at AD	POLE	125951.8N 0801104.5E	79 FT	NIL	LIGHT POLE
In circling area and at AD	POLE	125951.9N 0801105.3E	79 FT	NIL	LIGHT POLE
In circling area and at AD	TREE	125952.1N 0801105.8E	93 FT	NIL	GROUP OF TREES
In circling area and at AD	TREE	125957.3N 0801115.0E	141 FT	NIL	GROUP OF TREES
In circling area and at AD	OTHER	130003.8N 0801125.0E	166 FT	NIL	MAST
In circling area and at AD	BUILDING	130015.0N 0801131.0E	211 FT	NIL	SHED
In circling area and at AD	BUILDING	130012.4N 0801132.1E	203 FT	NIL	BUILDING
In circling area and at AD	BUILDING	130011.3N 0801133.1E	193 FT	NIL	BUILDING
In circling area and at AD	TREE	130016.1N 0801135.4E	269 FT	NIL	GROUP OF TREES
In circling area and at AD	TREE	130015.9N 0801136.3E	260 FT	NIL	GROUP OF TREES
In circling area and at AD	TREE	130018.3N 0801136.3E	308 FT	NIL	GROUP OF TREES
In circling area and at AD	BUILDING	130014.3N 0801136.4E	205 FT	NIL	BUILDING
In circling area and at AD	BUILDING	130039.5N 0800911.3E	96 FT	NIL	TANK ON BUILDING (BUILDING TOP - 27.3 M)
In circling area and at AD	BUILDING	130036.8N 0800916.1E	88 FT	NIL	BUILDING
In circling area and at AD	BUILDING	130035.6N 0800918.2E	86 FT	NIL	BUILDING
In circling area and at AD	TREE	130035.9N 0800919.8E	104 FT	NIL	Group of Trees
In circling area and at AD	TREE	130037.8N 0800915.0E	98 FT	NIL	Group of Trees

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	TREE	130033.3N 0800919.8E	85 FT	NIL	Group of Trees
In circling area and at AD	ANTENNA	130020.0N 0800939.8E	76 FT	NIL	CWIS ANTENNA
In circling area and at AD	OTHER	130028.9N 0800925.7E	57 FT	NIL	MOBILE ROAD TRAFFIC (ROAD ELEV. 12.3M + 5.0M TRAFFIC HEIGHT)
In circling area and at AD	BUILDING	130026.1N 0800928.8E	45 FT	NIL	BIRD SCARING
In circling area and at AD	BUILDING	130029.5N 0800925.0E	65 FT	NIL	SHED TOP
In circling area and at AD	TREE	130024.6N 0800906.5E	86 FT	NIL	GROUP OF TREES
In circling area and at AD	BUILDING	130034.8N 0800918.3E	87 FT	NIL	BUILDING
In circling area and at AD	BRIDGE	130036.8N 0800915.4E	99 FT	NIL	BUILDING
In circling area and at AD	TREE	130034.7N 0800919.1E	91 FT	NIL	Group of Trees
In circling area and at AD	TREE	130035.4N 0800917.5E	87 FT	NIL	BUILDING U/C
In circling area and at AD	TREE	130036.4N 0800915.8E	92 FT	NIL	GROUP OF TREES
In circling area and at AD	ANTENNA	125946.3N 0801042.8E	60 FT	NIL	MET ANTENNA
In circling area and at AD	POLE	125937.6N 0801108.1E	73 FT	NIL	ELECTRIC POLE
In circling area and at AD	TREE	125927.0N 0801100.2E	99 FT	NIL	GROUP OF TREES
In circling area and at AD	TREE	125937.9N 0801108.3E	83 FT	NIL	TREE
In circling area and at AD	TREE	125929.5N 0801054.7E	87 FT	NIL	TREE
In circling area and at AD	TREE	125928.7N 0801055.3E	104 FT	NIL	TREE
In circling area and at AD	TREE	125927.6N 0801057.0E	113 FT	NIL	GROUP OF TREES
In circling area and at AD	BUILDING	125930.2N 0801055.7E	84 FT	NIL	BUILDING
In circling area and at AD	TREE	125930.6N 0801055.4E	92 FT	NIL	TREE
In circling area and at AD	TANK	125925.3N 0801100.5E	124 FT	NIL	OVER HEAD WATER TANK
In circling area and at AD	TREE	125936.9N 0801110.1E	106 FT	NIL	TREE
In circling area and at AD	OTHER	125936.9N 0801110.8E	108 FT	NIL	CELL PHONE MAST ON BUILDING TOP
In circling area and at AD	TREE	125923.5N 0801104.8E	100 FT	NIL	TREE

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	125943.9N 0801034.8E	60 FT	NIL	WIND DIRECTION INDICATOR
In circling area and at AD	ANTENNA	125936.7N 0801111.4E	95 FT	NIL	ANTENNA ON BUILDING (LIGHTED TOP - 27.6 M, BUILDING TOP - 26.8 M)
In circling area and at AD	BUILDING	125932.7N 0801301.5E	206 FT	NIL	BUILDING
In circling area and at AD	BUILDING	125925.6N 0801259.9E	215 FT	NIL	BUILDING
In circling area and at AD	BUILDING	125929.0N 0801056.0E	102 FT	NIL	BUILDING
In circling area and at AD	BUILDING	125931.7N 0801054.0E	77 FT	NIL	VVIP CEREMONY LOUNGE & GENERAL AVIATION TERMINAL
In circling area and at AD	POLE	125930.8N 0801055.2E	83 FT	NIL	LIGHT POST
In circling area and at AD	OTHER	125938.2N 0801107.3E	75 FT	NIL	RED LIGHT ON ROAD
In circling area and at AD	TREE	125930.3N 0801055.5E	108 FT	NIL	Group of Trees
In circling area and at AD	BUILDING	125929.7N 0801056.9E	103 FT	NIL	Building
In circling area and at AD	BUILDING	125931.1N 0801053.4E	79 FT	NIL	VVIP CEREMONY LOUNGE & GENERAL AVIATION TERMINAL
In circling area and at AD	STACK	125936.4N 0801112.5E	96 FT	NIL	CHIMNEY
In circling area and at AD	BUILDING	125936.8N 0801114.1E	110 FT	NIL	BUILDING U/C
In circling area and at AD	BUILDING	125935.3N 0801115.8E	93 FT	NIL	MOSQUE
In circling area and at AD	BUILDING	125928.6N 0801056.8E	107 FT	NIL	BUILDING
In circling area and at AD	BUILDING	125929.9N 0801056.1E	97 FT	NIL	BUILDING
In circling area and at AD	OTHER	125937.2N 0801111.4E	108 FT	NIL	CELL PHONE MAST
In circling area and at AD	BUILDING	125928.1N 0801057.6E	102 FT	NIL	BUILDING
In circling area and at AD	BUILDING	125929.1N 0801057.7E	102 FT	NIL	BUILDING
In circling area and at AD	BUILDING	125927.6N 0801058.5E	94 FT	NIL	BUILDING
In circling area and at AD	BUILDING	125927.7N 0801059.8E	91 FT	NIL	Building
In circling area and at AD	BUILDING	125926.0N 0801101.3E	95 FT	NIL	Building

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	125926.2N 0801102.3E	94 FT	NIL	Building
In circling area and at AD	BUILDING	125925.2N 0801103.5E	90 FT	NIL	BUILDING
In circling area and at AD	BUILDING	125924.3N 0801104.5E	98 FT	NIL	BUILDING
In circling area and at AD	BUILDING	125924.0N 0801103.7E	98 FT	NIL	BUILDING
In circling area and at AD	TREE	125930.8N 0801056.1E	102 FT	NIL	GROUP OF TREES
In circling area and at AD	OTHER	125937.9N 0801107.2E	75 FT	NIL	RED LIGHT ON ROAD
In circling area and at AD	TREE	125925.6N 0801103.0E	114 FT	NIL	GROUP OF TREES
In circling area and at AD	TREE	125924.5N 0801101.1E	118 FT	NIL	GROUP OF TREES
In circling area and at AD	ANTENNA	125949.1N 0801042.0E	75 FT	NIL	CIWS ANTENNA
In circling area and at AD	BUILDING	125923.5N 0801103.9E	99 FT	NIL	BUILDING
In circling area and at AD	TREE	125937.6N 0801113.3E	110 FT	NIL	GROUP OF TREES
In circling area and at AD	OTHER	130023.0N 0800914.7E	58 FT	NIL	MOBILE ROAD TRAFFIC (ROAD ELEV . + TRAFFIC HEIGHT 12.7 + 5M)
In circling area and at AD	BUILDING	130026.3N 0800914.9E	53 FT	NIL	WATCH TOWER
In circling area and at AD	NAVAID	125909.8N 0800917.6E	56 FT	NIL	GP MONIITOR ANTENNA
In circling area and at AD	FENCE	130027.0N 0800918.0E	54 FT	NIL	AIRPORT BOUNDARY WALL WITH FENCING ON TOP
In circling area and at AD	OTHER	130026.7N 0800918.7E	56 FT	NIL	MOBILE ROAD TRAFFIC (ROAD ELEV. + TRAFFIC HEIGHT 12.1 + 5M)
In circling area and at AD	NAVAID	125910.7N 0800920.1E	92 FT	NIL	GP ANTENNA
In circling area and at AD	NAVAID	125910.8N 0800920.1E	74 FT	NIL	DVOR MONITOR ANTENNA
In circling area and at AD	OTHER	125913.6N 0800922.2E	67 FT	NIL	WDI
In circling area and at AD	OTHER	125911.2N 0800922.5E	49 FT	NIL	MET INSTRUMENT
In circling area and at AD	FENCE	130028.8N 0800924.4E	52 FT	NIL	AIRPORT BOUNDARY WALL WITH FENCING ON TOP

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	130028.5N 0800925.2E	57 FT	NIL	MOBILE ROAD TRAFFIC (ROAD ELEV. + TRAFFIC HEIGHT 12.4 + 5M)
In circling area and at AD	OTHER	125904.1N 0800937.8E	103 FT	NIL	FLOOD LIGHT
In circling area and at AD	OTHER	125907.2N 0800946.6E	103 FT	NIL	FLOOD LIGHT
In circling area and at AD	OTHER	125907.5N 0800947.6E	104 FT	NIL	FLOOD LIGHT
In circling area and at AD	ANTENNA	130014.8N 0800954.6E	64 FT	NIL	ANTENNA
In circling area and at AD	BUILDING	130013.0N 0800954.9E	45 FT	NIL	BIRD CHASING HUT
In circling area and at AD	OTHER	130011.2N 0800956.0E	65 FT	NIL	WDI
In circling area and at AD	OTHER	125955.4N 0800959.5E	104 FT	NIL	FLOOD LIGHT
In circling area and at AD	BUILDING	130002.9N 0801011.0E	39 FT	NIL	BIRD CHASING HUT
In circling area and at AD	ANTENNA	125914.8N 0801011.3E	149 FT	NIL	ANTENNA ON BUILDING TOP
In circling area and at AD	BUILDING	125916.1N 0801012.0E	198 FT	NIL	ATC BUILDING LIGHTNING ARRESTOR (BUILDING TOP - 56.1 m)
In circling area and at AD	OTHER	125916.2N 0801012.2E	186 FT	NIL	ATC BEACON
In circling area and at AD	OTHER	125932.2N 0801052.3E	64 FT	NIL	MOBILE ROAD TRAFFIC(ROAD ELEV . + TRAFFIC HEIGHT 14.5 + 5M)
In circling area and at AD	NAVAID	125944.9N 0801051.2E	103 FT	NIL	GP MAIN ANTENNA L/A TOP (LIGHTED TOP- 31.2 M.)
In circling area and at AD	NAVAID	125945.0N 0801051.2E	69 FT	NIL	GP DME ANTENNA L/A TOP (LIGHTED TOP - 20.6M.)
In circling area and at AD	NAVAID	125946.0N 0801054.1E	68 FT	NIL	GP MONITOR ANTENNA L/A TOP (ANTENNA TOP-20.6M.)
In circling area and at AD	BUILDING	125945.0N 0801050.9E	61 FT	NIL	GP HUT
In circling area and at AD	NATURAL HIGHPOINT	125343.9N 0800542.2E	564 FT	NIL	HILL TOP
In circling area and at AD	OTHER	125723.5N 0800910.2E	236 FT	NIL	LIGHTNING ARRESTOR ON BUILDING (BUILDING TOP - 66.4 M)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM 08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	125719.6N 0800906.7E	253 FT	NIL	LIGHTNING ARRESTOR ON SKYPARK RESIDENCE (BUILDING TOP - 72.8 M)
In circling area and at AD	OTHER	125725.3N 0800912.9E	238 FT	NIL	LIGHTNING ARRESTOR (BUILDING TOP - 66.8 M)
In circling area and at AD	BUILDING	125817.7N 0800937.5E	531 FT	NIL	TEMPLE ON HILL TOP
In circling area and at AD	OTHER	125815.9N 0800940.1E	533 FT	NIL	MAST ON HILL TOP
In circling area and at AD	OTHER	125832.6N 0800951.8E	360 FT	NIL	STRUCTURE ON HILL TOP
In circling area and at AD	OTHER	125830.1N 0800953.4E	406 FT	NIL	CRANE ON HILL TOP
In circling area and at AD	NATURAL HIGHPOINT	125834.8N 0800959.0E	447 FT	NIL	HILL TOP
In circling area and at AD	TOWER	130400.5N 0801651.5E	613 FT	NIL	TV TOWER
In circling area and at AD	OTHER	130216.9N 0800853.4E	227 FT	NIL	LIGHTNING ARRESTOR ON OSIAN CHLOROPHY 2 (LIGHTED TOP - 69.4 M, BUILDING TOP - 66.9 M)
In circling area and at AD	OTHER	130219.1N 0800853.1E	227 FT	NIL	LIGHTNING ARRESTOR ON OSIAN CHLOROPHY 1(LIGHTED TOP - 69.1 M, BUILDING TOP - 67.2 M)
In circling area and at AD	ANTENNA	125916.2N 0801012.3E	188 FT	NIL	ANTENNA ON ATC BUILDING
In circling area and at AD	OTHER	130018.8N 0801138.2E	307 FT	NIL	CHURCH POLE

VOMM AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

1	Name of the associated meteorological office	Chennai
2	Hours of service and, where applicable, the designation of the responsible meteorological office outside these hours	H24
3	Office responsible for preparation of TAFs and periods of validity and interval of issuance of the forecasts	Chennai TAFs 9 hourly valid 0000-0900, 0300-1200, 0600-1500, 0900-1800, 1200-2100, 1500-2400, 1800-0300 & 2100-0600 UTC and 30 hourly TAFs valid 0000-0600 of next day, 0600-1200 of next day, 1200-1800 of next day & 1800-2400 UTC of next day
4	Availability of the trend forecast for the aerodrome and interval of issuance	Trend Trend forecast appended in each METAR at 30 min and SPECI when required

5	Information on how briefing and/or consultation is provided	Charts in tabular form, Manual or through on-line briefing system.
6	Types of flight documentation supplied and language(s) used in flight documentation	Tabular & Chart Form (English)
7	Charts and other information displayed or available for briefing or consultation	1. WAFC products: Data for upper wind & temperature for levels : 850, 700, 500, 300, 250, 200, 150 hPa. SIG WX charts for FL 250-630. 2. National charts for consultation only: Surface charts, synoptic charts, prognostic upper air wind and temperature charts for levels : 850, 700, 600, 500, 4000, 300, 250, 200, 150, 100 hPa, Prognostic upper air charts at constant pressure, Prognostic significant weather chart, DWR products / satellite imagery.
8	Supplementary equipment available for providing information on meteorological conditions, e.g. weather radar and receiver for satellite images;	Metar/Warning Display System Online Briefing system, DCWIS, Transmissometer (Drishty)
9	The air traffic services unit(s) provided with meteorological information	VOMM Chennai ATC, ACC and FIC
10	Additional information, e.g. concerning any limitation of service.	NIL

VOMM AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

Designations	TRUE Bearings	Dimensions of RWY (M)	Strength of pavement (PCN) and associated data) and surface of runway and associated stopways	Geographical coordinates for threshold and runway end
1	**2**	**3**	**4**	**5**
12	117.65 DEG	2890 x 45 M	89/F/C/W/T Concrete/Asphalt	THR: 130013.15N 0800944.95E RWY END: 125941.28N 0801046.95E
30	297.65 DEG	2890 x 45 M	89/F/C/W/T Concrete/Asphalt	THR: 125941.28N 0801046.95E RWY END: 130021.76N 0800928.16E
07	69.05 DEG	3661 x 45 M	105/F/C/W/T Concrete/Asphalt	THR: 125902.93N 0800910.58E RWY END: 125945.52N 0801104.02E
25	249.05 DEG	3661 x 45 M	105/F/C/W/T Concrete/Asphalt	THR: 125945.52N 0801104.02E RWY END: 125902.93N 0800910.58E

THR elevation and highest elevation of TDZ of precision APP RWY	Slope of runway and associated stopway	Dimensions of stopway (M)	Dimensions of clearway (M)	Dimensions of strips (M)
6	7	8	9	10
THR: 44.0FT TDZ: 44.0FT	0.13%	NIL	NIL	3010 x 280 M
THR: 49.0FT TDZ: 48.0FT	0.13%	NIL	NIL	3010 x 280 M
THR: 43.0FT TDZ: 43.0FT	0.01%	50 x 45 M	153 x 300	3891 x 280 M
THR: 54.0FT TDZ: 54.0FT	0.01%	60 x 45 M	205 x 300	3891 x 280 M

Dimensions of runway end safety areas	Location and description of arresting system (if any)	Existence of an obstacle-free zone	Remarks.
11	12	13	14
240M x 90MPaved		AVBL	1. PCN: From beginning of RWY 12 to 960M : 85/R/B/W/T From 960M of RWY 12 to 1010M : 89/F/C/W/T From 1010M of RWY 12 to 1560M : 98/F/A/W/T From 1560M of RWY 12 to 2890 M : 89/F/C/W/T 2. Vertical Datum: EGM08
210M x 90MPaved		AVBL	1. PCN: From beginning of RWY 12 to 960M : 85/R/B/W/T From 960M of RWY 12 to 1010M : 89/F/C/W/T From 1010M of RWY 12 to 1560M : 98/F/A/W/T From 1560M of RWY 12 to 2890 M : 89/F/C/W/T 2. Vertical Datum: EGM08
240M x 90M		AVBL	1. Vertical Datum: EGM08
90M x 90M		AVBL	1. Vertical Datum: EGM08

VOMM AD 2.13 DECLARED DISTANCES

RWY Designator	Take-off run available TORA (M)	Take-off distance available TODA (M)	Accelerate distance available ASDA (M)	Landing distance available LDA (M)	Remarks (including runway entry or start point where alternative reduced declared distances have been declared)
1	2	3	4	5	6
12	2890	2890	2890	2110	Threshold displaced by 780M
30	2680	2680	2680	2680	

RWY Designator	Take-off run available TORA (M)	Take-off distance available TODA (M)	Accelerate distance available ASDA (M)	Landing distance available LDA (M)	Remarks (including runway entry or start point where alternative reduced declared distances have been declared)
1	2	3	4	5	6
07	3661	3814	3711	3661	
25	3661	3866	3721	3661	

VOMM AD 2.14 APPROACH AND RUNWAY LIGHTING

Runway Designator	Type, length and intensity of approach lighting system	Runway threshold lights, colour and wing bars	Type of visual slope indicator system	Length of runway touchdown zone lights
1	2	3	4	5
12	NIL	Green	PAPI LEFT/3.00 DEG	NIL
30	SALS 300 M LIH	Green	PAPI LEFT/3.00 DEG	NIL
07	CAT I 900 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (74.44FT)	NIL
25	CAT I 510 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (84.58FT)	900 M

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
NIL	2890 M 60 M White LIH	Red	NIL	i) RWY Edge Lights Variable white, except that llights between the beginning of the runway and the 780M displaced threshold show red in the approach direction. Last 600M of RWY Edge light are yellow in approach direction.
NIL	2890 M 60 M White LIH	Red	NIL	i) RWY Edge Lights Variable white. Last 600M of RWY Edge light are yellow in approach direction.

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
3661 M 30 M White	3661 M 60 M White LIH	Red	Red 50 M	i) Runway Centreline Lights: Last 300 M: Red Between last 300 M to 900 M in the approach direction: Alternate Red ii) RWY Edge Lights Variable white. Last 600M of RWY Edge light are yellow in approach direction.
3661 M 30 M White	3661 M 60 M White LIH	Red	Red 60 M	i) Runway Centreline Lights: Last 300 M: Red Between last 300 M to 900 M in the approach direction: Alternate Red ii) RWY Edge Lights Variable white. Last 600M of RWY Edge light are yellow in approach direction.

VOMM AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

1	Location, characteristics and hours of operation of aerodrome beacon/identification beacon (if any)	ABN	At Tower Building, FLG W&G EV 2SEC, H24
		IBN	NIL
2	Location and lighting (if any) of anemometer/ landing direction indicator;	LDI	LDI provided at south side of RWY 07/25 abeam ATC Tower.
		Anemometer	Near RWY 07 & RWY25, Lighted
3	Taxiway edge and taxiway centre line lights;	Edge	All TWY
		Centre Line	---
4	Secondary power supply including switch-over time;	Secondary Power Supply (DG Supply) have been provided to all Visual and Non-Visual Aids. All Visual Aids of RWY is provided with further UPS Backup with Switch Over Time of 1 Second. All Taxiways and Guard Lights are provided with Power Supply of Switch Over Time of 15 Seconds.	
5	Remarks	NIL	

VOMM AD 2.16 HELICOPTER LANDING AREA

1	Geographical coordinates of the geometric centre of touchdown and lift-off (TLOF) or of each threshold of final approach and take-off (FATO) area	Not Established
2	TLOF and/or FATO area elevation:	Not Established
3	TLOF and FATO area dimensions to the nearest metre or foot, surface type, bearing strength and marking;	Not Established
4	True bearings of FATO;	Not Established

5	Declared distances available	Not Established
6	Approach and FATO lighting;	Not Established
7	Remarks	Not Established

VOMM AD 2.17 AIR TRAFFIC SERVICE AIRSPACE

1	Airspace designation, geographical coordinates and lateral limits	CTR: Circular area centered on 125915N 0800918E within a 25NM radius excluding the lateral and vertical limit of Arrakonam MCTR-A.
2	Vertical limits	FL 55
3	Airspace classification	D
4	Call sign and language(s) of the air traffic services unit providing service;	Chennai Tower / Chennai Approach, English
5	Transition altitude	4000 FT
6	Hours of applicability	
7	Remarks	NIL

VOMM AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

Service Designation	Call sign	Channel(s)	SATVOICE Number(s), if available
1	2	3	4
OTHER	CHENNAI CLEARANCE DELIVERY	121.625 MHZ	
OTHER	CHENNAI RADIO	3470.000 KHZ	
OTHER	CHENNAI RADIO	5670.000 KHZ	
OTHER	CHENNAI RADIO	6583.000 KHZ	
OTHER	CHENNAI RADIO	8861.000 KHZ	
OTHER	CHENNAI RADIO	8909.000 KHZ	
OTHER	CHENNAI RADIO	11285.000 KHZ	
OTHER	CHENNAI RADIO	13318.000 KHZ	
SAR	-	123.100 MHZ	SHORT CODE: 441930 PSTN: 8991117568
SAR	-	3023.000 KHZ	
SAR	-	5680.000 KHZ	
APP	CHENNAI APPROACH	125.500 MHZ	
APP	CHENNAI APPROACH ARRIVAL	120.750 MHZ	
APP	CHENNAI APPROACH DEPARTURE	127.900 MHZ	
TWR	CHENNAI TOWER	118.100 MHZ	
TWR	CHENNAI TOWER	118.800 MHZ	
ATIS	CHENNAI INFORMATION	127.450 MHZ	
ACC	CHENNAI CONTROL	118.900 MHZ	
ACC	CHENNAI CONTROL	119.250 MHZ	
ACC	CHENNAI CONTROL	124.450 MHZ	
ACC	CHENNAI CONTROL	124.500 MHZ	
ACC	CHENNAI CONTROL	125.300 MHZ	

Service Designation	Call sign	Channel(s)	SATVOICE Number(s), if available
ACC	CHENNAI CONTROL	126.150 MHZ	SHORT CODE: 441933 PSTN: 8991117569
ACC	CHENNAI CONTROL	128.100 MHZ	
ACC	CHENNAI CONTROL	128.700 MHZ	
ACC	CHENNAI CONTROL	132.250 MHZ	
ACC	CHENNAI CONTROL	132.500 MHZ	
ACC	CHENNAI CONTROL	132.950 MHZ	
ACC	CHENNAI CONTROL	133.950 MHZ	
ACC	CHENNAI CONTROL	134.250 MHZ	
ACC	CHENNAI CONTROL	134.300 MHZ	
ACC	CHENNAI CONTROL	134.425 MHZ	
ALRS	---	121.500 MHZ	
SMC	CHENNAI GROUND	121.900 MHZ	

Logon address, as appropriate	Hours of operation	Remarks
5	6	7
	H24	CLD
	H24	RADIO
	H24	RADIO
	H24	RADIO
	H24	RADIO
	H24	RADIO
	H24	RADIO
	H24	RADIO
	H24	NIL
	H24	RADIO
	H24	RADIO
	H24	BACKUP
	H24	APPROACH ARRIVAL
	H24	APPROACH DEPARTURE
	H24	NIL
	H24	BACKUP
	H24	NIL
	H24	UBE
	H24	BACKUP
	NIL	BACKUP
	H24	UBW
	H24	UMM-S
	H24	NIL
	H24	UHS
	H24	BACKUP
	H24	BACKUP
	H24	UTV

Logon address, as appropriate	Hours of operation	Remarks
	NIL	NIL
	H24	BACKUP
	H24	UMM-N
	H24	BACKUP
	H24	UML
	H24	NIL
	H24	NIL

VOMM AD 2.19 RADIO NAVIGATION AND LANDING AIDS

Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid	Identification	Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate	Hours of operation, as appropriate;
1	2	3	4
LOC 07 CAT I	IMAS	110.300 MHz	H24
LOC 25 CAT I	ICHN	109.700 MHz	H24
GP 07		335.000 MHz	H24
GP 25	ICHN	333.200 MHz	H24
DME ILS 07	IMAS	CH40X	H24
DME ILS 25	ICHN	CH34X	H24
DVOR/DME	MMV	112.500 MHz CH72X	H24
DVOR/DME	CNI	114.900 MHz CH96X	

Geographical coordinates of the position of the transmitting antenna	Elevation of transmitting antenna of DME/ elevation of GBAS reference point	Service volume radius from the GBAS reference point	Remarks
5	6	7	8
125950.1N 0801116.1E			
125859.3N 0800900.8E			
125910.7N 0800920.1E			
125944.9N 0080151.2E			Glide angle 3 DEG
125910.9N 0800920.0E			
125945.0N 0801051.2E	68 FT		Vertical Datum: EGM08
125915.5N 0800918.1E	70 FT		1. Vertical datum: EGM08
130015.5N 0800958.0E	72 FT		1. Vertical datum: EGM08

VOMM AD 2.20 LOCAL AERODROME REGULATIONS

1. To reduce RT congestion and RWY occupancy time, following local regulations need to be followed:

i. Preferred exit TWY to vacate RWY 07: TWY Z for aircraft upto Code C. TWY Y for aircraft Code D and above

ii. Preferred exit TWY to vacate RWY 25:TWY V for aircraft up to Code C. TWY M for aircraft Code D and above - Exit speed maximum 40 KT.

If unable, to inform ATC as early as possible.

iii. All departures for RWY07 entering via TWY K to depart from TWY K intersection without backtracking on the RWY. (Take-off run available from K intersection is 3580 m). Aircraft requiring backtrack for full length departure shall inform ATC prior to startup request.

iv. Departing aircraft passing 1000 ft to change over to Chennai APP (Primary frequency 127.900 MHz), unless otherwise specified by ATC tower.

v. Departing aircraft, on first contact with APP, shall transmit only the call sign, altitude passing and the assigned SID. In all other cases of first contact with an APP unit in Chennai, aircraft shall transmit only its call sign.

vi. ACARS equipped aircraft to obtain pre-departure clearance through datalink

vii. Arriving aircraft after vacating RWY to contact SMC Control (Chennai Ground, frequency: 121.900 MHz) unless otherwise instructed by tower.

viii. Arriving aircraft on reaching the allotted stand, need not contact ATC for informing the same.

ix. Aircraft shall comply with speed restrictions whenever imposed by ATC.

x. Aircraft to normally execute rate one turns as far as practicable.

xi. Arriving and departing aircraft to minimize RWY occupancy timefor enhancing RWY capacity.

xii. Aircraft not to request time to touch down and direct routing. ATC whenever practicable will issue the same.

2. Code C departure from RWY25 shall use TWY C or TWY N5 for line-up and Take-off as per ATC instruction. Aircraft requiring other than TWY C or TWY N5 shall inform ATC prior to start up.

3. 180 DEG turn on the runway Is not permitted for aircraft code C and above. Departure from extremity of RWY 07 only via TWY K and RWY 25 only via TWY J.

4. No Aircraft shall take lock turn/ pivotal turn on the Runway to avoid pavement damage.

6. AIRPORT COLLABORATIVE DECISION MAKING (A-CDM)

6.1 INTRODUCTION

6.1.1 Efficiency of Air Transport System is highly dependent on traffic predictability. A- CDM effectively enhances predictability (this reduces buffer times for resource planning and flight times), overall efficiency and punctuality by linking and sharing of accurate and timely information amongst Aircraft Operators, Airport Operator, ATC etc.

6.1.2 A-CDM is a Common Information Sharing Platform introduced to process timely and accurate information for quick and precise decision making. A-CDM takes into consideration the ETAs of arriving aircraft, Target Off Block Times of departures, Runway-in-use and the handling capacity of the runway and determines the arrival and departure sequence. Based on the sequence and the parking position, it calculates the Target Take-Off Times & Target Start up Times of departures and Target In-Block Times for arrivals.

6.2 TWO KEY PARAMETERS OF A-CDM: TOBT and TSAT

6.2.1 TARGET OFF-BLOCK TIME (TOBT)

6.2.1.1 Definition: The time that an Aircraft Operator estimates that an aircraft will be ready, all doors closed, boarding bridge removed, push back vehicle available and ready to start up/push back immediately upon reception of clearance from the SMC (GROUND). It is the real time target of ensuring readiness of an aircraft and therefore, more accurate than the static departure time mentioned in the flight plan and hence, an ideal milestone to be used by all airport partners.

6.2.1.2 Communication of TOBT:

i. If no delay is expected, EOBT will be treated as TOBT. Action need not be performed by the Aircraft Operator in the ACDM interface.

ii.For delay within 30 minutes of the EOBT, TOBT shall be updated in the ACDM interface by the aircraft operator. Optionally, EOBT may be revised via AFTN through DLA/CHG message.

iii. For delay more than 30 minutes from the EOBT, EOBT shall be mandatorily revised through a DLA/CHG message. (Option for changing TOBT will not be available in this case).

6.2.1.3 Accuracy: TOBTs must be updated to an accuracy of +/- 5 minutes.

6.2.2 TARGET START UP APPROVAL TIME (TSAT)

6.2.2.1 Definition: The time calculated by A-CDM application taking into account TOBT, CTOT and/or the traffic situation that an aircraft can expect to start-up/push back for departure.

Note: In case there is no congestion, ATC may allow the aircraft actual start up approval (ASAT) even before the TSAT.

6.2.2.2 Calculation of TSAT: TSAT is calculated automatically by A-CDM Application. The TSAT is calculated by taking into account TOBT, priority of aircraft, variable taxi times, and the runway capacity. Any revision of EOBT will trigger the A-CDM application to recalculate a new TSAT for that flight as well as to re-allocate earlier TSAT to next flight who meets the requisite conditions. TSAT is displayed in various ACDM interfaces in the following colour coded information.

•Blue colour: TSAT -15 minutes up to TSAT -5 minutes

•Green colour: TSAT -5 minutes up to TSAT +5 minutes

•Red colour: After TSAT +5 minutes.

6.2.2.3 Generation of TSAT: TSAT is generated by ACDM interface at TOBT – 60 minutes.

6.2.2.4 Communication of TSAT: TSAT will be displayed in various A-CDM interfaces. The AO/GHA shall advise flight crew of TSAT, displayed in the A-CDM interface.

6.2.2.5 Priorities for generation of TSAT:

The following aircraft will be accorded priority by A-CDM application at the time of generating TSATs:

i.VVIP flights

ii.Ambulance aircraft

iii.Aircraft in relief missions

iv.Aircraft returning to stand after push back/taxiing out for any reason.

6.2.2.6 Airlines/GHAs are required to monitor the A-CDM interface regularly to get information on the revised TSAT if any, in respect of their aircraft. It may please be noted that the earlier allocated TSAT may get changed due to the following reasons:

i.Change of runway.

ii.Change in taxi times.

iii.Revised TOBT of the subject aircraft.

iv.Cancellation/ revision in TOBT of other aircraft in the departure sequence.

v.Priority handling.

vi.Unusual Occurrences, etc.

vii.Application of ATFM regulation or ATC restriction resulting in new CTOT or start up delay.

viii.Inclement Weather.

ix.Bird activity.*

x.Change of route (SID)*

* [Future provision]

6.2.2.7 Accuracy: The TSAT will remain valid for +/- 5 minutes.

6.3 Co-ordination with the Air Traffic Flow Management Unit (ATFMU) *

6.3.1 A fully automatic data exchange with the ATFMU is established. A-CDM system is receiving ELDT and CTOT from C-ATFM.

6.3.2 The data sent from A-CDM to C-ATFM enables efficient CTOT calculations. ALDT, AIBT, TOBT, TSAT, TTOT, AOBT, ATOT, Registration and Type of Aircraft are sent.

6.4 A-CDM Start-up/Push-back procedures

6.4.1 Pre-Departure Sequence: Pre-Departure sequence is the order, in which the ATC plans the aircraft to depart from their Gate/Stands. It should not be confused with the Pre- Take-off sequence where ATC decides on the order in which the aircraft at holding points of the runway will depart.

6.4.2 Start-Up and Push-Back Procedure:

i.The aircraft must be ready for Start-up/Push-back at TOBT. Any revisions in TOBT shall be updated by the aircraft operator at the earliest.

ii. The Pilot will contact Clearance Delivery (CLD) and request for Pre-Departure Clearance between TSAT -15 minutes to TSAT-5 minutes. (Blue Zone)

iii. Pilot shall contact SMC (GROUND) for Start-up/Push-back between TSAT-5 minutes and TSAT+ 5 minutes (Green Zone), subject to change over from CLD.

Note: The system may prepone TSAT up to TOBT

iv. If by TSAT +5 minutes (Red Zone after TSAT+5 minutes), ATC has not received Start-up/Push-back request, the aircraft will lose its position in sequence. In such case:

a. Aircraft Operator shall update the TOBT.b. A revised TSAT will be issued because of resequencing done based on new TOBT.

v. ATC should normally be able to issue start-up/push-back at TSAT. Pilots will be informed by ATC if there is a delay.

vi. Taxi clearance shall be requested within 5 minutes of Start-up/Push-back approval time.

Note: If this has not occurred, SMC (GROUND) must be notified of the extent of delay. In such cases, aircraft may lose its departure slot and a new TOBT may be required. Aircraft Operator shall be responsible to obtain new ADC number from IAF-MLU.

vii. The Pre-Departure (Off- block) Sequence will be determined in accordance with Target Start up Approval Time (TSAT) and NOT in accordance with the Start-up Request. Pre-Departure Sequence will not have any bearing on Actual Departure sequence.

(Note: Actual Departure sequence may differ from pre-departure sequence in order to optimize the Runway and Airspace utilization.)

6.5 INFORMATION DISSEMINATION TO A-CDM STAKEHOLDERS

Relevant information for stakeholders, available with A-CDM system, is disseminated through following means:

i. Access controlled application interface provided to representatives of Airline Operators, Ground Handlers and Airport Operator.

ii. Web interface that can be accessed over Internet.*

iii. Display boards on the airside at certain parking stands, in the viewable area of cockpit crew, where limited information is disseminated.*

* [Future provision]

6.6 TERMS AND ABBREVIATIONS

All Acronyms are time parameters which have a standard length of four characters and have been arranged as per aircraft movement sequence.

[While developing the A-CDM procedures, the guidance material published by Euro control, ICAO and other Organizations has been used].

S. No.	Acronyms	Definition	Explanation
6.1	ELDT	Estimated Landing Time	The estimated time that an aircraft will touchdown on the runway. (Equivalent to ATC ETA = Estimated Time of Arrival = landing).
6.2	ALDT	Actual Landing Time	The time that an aircraft lands on a runway. (Equivalent to ATC ATA = Actual Time of Arrival = Landing, ACARS = ON).
6.3	EXIT	Estimated Taxi-In Time	The estimated taxi time between landing and in- Block.
6.4	AXIT	Actual Taxi-In Time	AXIT = AIBT – ALDT
6.5	SIBT	Scheduled In- Block Time	The time that an aircraft is scheduled to arrive at its first parking position.
6.6	EIBT	Estimated In- Block Time	The estimated time that an aircraft will arrive in- blocks. (Equivalent to Aircraft Operator's ETA = Estimated Time of Arrival).
6.7	AIBT	Actual In- Block Time	The time that an aircraft arrives in-blocks. (Equivalent to Aircraft Operator's ATA = Actual Time of Arrival, ACARS = IN).
6.8	STTT	Scheduled Turnaround Time	STTT = SOBT – SIBT
6.9	ETTT	Estimated Turnaround Time	The time estimated by the AO/GH on the day of operation to turnaround a flight taking into account the operational constraints.
6.10	MTTT	Minimum Turnaround Time	The minimum turnaround time agreed with an AO/GH for a specified flight or aircraft type.
6.11	ATTT	Actual Turnaround Time	ATTT = AOBT – AIBT
6.12	ASRT	Actual Start Up Request Time	Time the pilot requests start up clearance.
6.13	TSAT	Target Start Up Approval Time	The time provided by ATC taking into account TOBT, CTOT* and/or the traffic situation that an Aircraft can expect start-up / push back approval. Note: The actual start up approval (ASAT) can be given in advance of TSAT.
6.14	ASAT	Actual Start Up Approval Time	Time that an aircraft receives its start-up approval.
6.15	SOBT	Scheduled Off- Block Time	The time that an aircraft is scheduled to depart from its parking position.
6.16	EOBT	Estimated Off- Block Time	The estimated time at which the aircraft will start movement associated with departure (ICAO).
6.17	TOBT	Target Off- Block Time	The time that an Aircraft Operator estimates that an aircraft will be ready, all doors closed, boarding bridge removed, push back vehicle available and ready to start up / push back immediately upon reception of clearance from the SMC (GROUND).
6.18	AOBT	Actual Off- Block Time	Time the aircraft pushes back / vacates the parking position. (Equivalent to Aircraft Operator's ATD = Actual Time of Departure & ACARS = OUT)
6.19	ARDT	Actual Ready Time (for Movement)	When the aircraft is ready for start-up/push back or taxi immediately after clearance delivery, meeting the requirements set by the TOBT definition.
6.20	EXOT	Estimated Taxi-Out Time	The estimated taxi time between off-block and take off. This estimate includes any delay buffer time at the holding point prior to take off.
6.21	AXOT	Actual Taxi- Out Time	AXOT = ATOT – AOBT
6.22	ETOT	Estimated Take Off Time	The estimated take off time taking into account the EOBT plus EXOT.

6.23	CTOT	Calculated Take Off Time	A time calculated and issued by the appropriate Central Management unit, as a result of tactical slot allocation, at which a flight is expected to become airborne.
6.24	TTOT	Target Take Off Time	The Target Take Off Time taking into account the TOBT/TSAT plus the EXOT. Each TTOT on one runway is separated from other TTOT or TLDT to represent vortex and/or SID separation between Aircraft.
6.25	ATOT	Actual Take Off Time	The time that an aircraft takes off from the runway. (Equivalent to ATC ATD = Actual Time of Departure, ACARS = OFF).

VOMM AD 2.21 NOISE ABATEMENT PROCEDURES

1. Consistent with safety of aircraft operations and in consideration of high intensity runway operations, pilots should minimize the use of reverse thrust after landing to reduce disturbance in areas adjacent to the aerodrome.

VOMM AD 2.22 FLIGHT PROCEDURES

I. TRANSPONDER OPERATING PROCEDURES ON GROUND

1. **Introduction:**

Advanced Surface Movement Guidance and Control System (A-SMGCS) using Mode-S Multi-lateration has been commissioned at Chennai International Airports.

The Aircraft Transponder Operating Procedures, particularly in the movement area of the airport(s), where A-SMGCS has been commissioned, is as given below:

2. **DEPARTURE**

i. At the Gate/Stand:
Select STBY
Enter the discrete SSR code received from Clearance Delivery/Surface Movement Control. Enter the three letter ICAO designator followed by the flight identification number (e.g. AIC748) through the FMS or the Transponder control panel, depending on the avionics.

ii. On requesting Pushback/Taxi (whichever is earlier):
Select Transponder or equivalent and AUTO if available
This action will enable the aircraft ID, used as the Call sign by ATC, to be displayed on the surveillance display of ATC. ATC can verify the data and use it for necessary identification.

iii. When Lining up:
Select TCAS
Select TCAS only after receiving the clearance to line up, to ensure that the performance of systems based on SSR frequencies (including airborne TCAS units, SSR and A-SMGCS) is not compromised.

3. **ARRIVAL**

i. When on the Runway:
Keep TCAS selected

ii. After vacating the Runway:
Select Transponder or equivalent and AUTO if available
There is a need that the Transponder remains able to exchange data with the A-SMGCS system. However to ensure that the performance of systems based on SSR frequencies (including airborne TCAS Unit, SSR & A-SMGCS) is not compromised, TCAS shall be deselected when vacating the Runway.

iii. Parked on Stand:
Select STBY
The Transponder will not reply to interrogation. The discrete SSR Code given to that particular flight can now be recycled for other flights.

Note:When on ground the aircraft must squawk Mode C, in order to provide the altitude information to the surveillance system, and thereby prevent
a. clutter on Terminal Approach Radar Display (and)
b. false automatic detection of departure for aircraft still on ground.

II. CHENNAI TMA ROUTING

Route Designator	Chennai TMA Routing
N571 (From West)	DORAM (N154304.3 E0772055.8) - 138º/232NM - CHENNAI MMV VOR
N571 (From East)	IDASO - (N123935.3 E0833323) - 278º/199NM CHENNAI MMV VOR

III. **ATS Surveillance Services**.

1 (a) : ADS-B based ATS surveillance services will be provided on opportunity basis on VHF 132.950 MHZ (Main)/124.450 MHZ (Backup) within the portion of airspace bounded by coordinates:
081234N0942500E-085553N0922347E, 122308N0890452E and then along 220NM Clockwise ARC centered at Sri Vijaya Puram DVOR/DME PPB to 142804N0901943E, 140000N0920000E-133000N0942500E to the point of origin.

1 (b) : ATS in the remaining airspace of Chennai Control Area- Sector UPB will be provided by Chennai Oceanic Control Unit using HF and CPDLC
FL 155/FL 460

2. ADS-B equipped aircraft flying within the jurisdiction of Chennai Oceanic Control shall comply with relevant provisions

IV. LOW VISIBILITY PROCEDURES

1. BACKGROUND:

In compliance with DGCA Civil Aviation Requirements (CAR) Section 8, Series 'C', Part-I on All-weather Operations, Low visibility procedures are to be enforced at aerodromes for ensuring safe operations during Categories II and III approaches and/or low visibility take-offs (sp. Also in the Revision 10 to the CAR para 5.3 following provision regarding Low Visibility Procedures is added. "Quote" ***An operator shall not conduct take-off with RVR/visibility less than standard Category I conditions of 550M RVR/800M visibility unless low visibility procedures are enforced.*** "Unquote".

This provision necessitated the establishment of Low Visibility Procedures for accommodating/permitting departures in Visibility/RVR less than 800M/550M even at such airports where there are no CAT II and CAT III operations.

At Chennai Airport, ILS Category I is available, accordingly, Low visibility Procedures have been developed for Chennai International Airport to accommodate/permit departures even in Visibility/RVR less than 800M/550M from RWY 07 & RWY 25 up to RVR of 200M and from RWY 12 & RWY 30 up to RVR of 300M, subject to availability of required facilities and manual RVR observations whenever instrument RVR is not available. Low Visibility Take off may be conducted on RWY07/25 up to RVR 150M when MID and END RVR observations are also made available.

2. DEFINITIONS:

2.1 Low Visibility Procedures (LVP): Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II and III approaches and/or low visibility take-offs.

Note: as per Para 5.3 of CAR on All Weather Operations, an operator shall not conduct Take-off with RVR/Visibility less than standard CAT-I conditions of 550M RVR/800M Visibility unless low visibility procedures are enforced.

2.2 Low Visibility Take Off (LVTO): A term used in relation to flight operations referring to a take-off on a runway where the RVR is less than 400 m.

2.3 Manoeuvring Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, excluding aprons.

2.4 Runway Visual Range: The range over which the pilot of an aircraft on the centreline of a runway can see the runway surface markings or the lights delineating the runway or identifying its centreline.

2.5 Aerodrome Operating Minima: The limits of usability of an aerodrome for:

a. Take off, expressed in terms of runway visual range and / or visibility and, if necessary, cloud conditions.
b. Landing in 2D instrument approach operations, expressed in terms of visibility and/or runway visual range; minimum descent altitude/height (MDA/H) and, if necessary, cloud conditions; and
c. Landing in 3D instrument approach operations, expressed in terms of visibilityand/or runway visual range and decision altitude/height (DA/H) appropriate to the type and/or category of the operation.

2.6 Visibility - Visibility for aeronautical purposes is the greater of:

a. The greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognized when observed against a bright background;
b. The greatest distance at which lights in the vicinity of 1000 candelas can be seen and identified against an unlit background.

Note 1: The two distances have different values in air of a given extinction coefficient, and the latter b) varies with the background illumination. The former a) is represented by the meteorological optical range (MOR).

Note 2: The definition applies to the observations of visibility in local routine and special reports, to the observations of prevailing and minimum visibility reported in METAR and SPECI and to the observations of ground visibility.

3. GENERAL

The Low Visibility Procedure (LVP) incorporates safeguarding measures to mitigate runway incursions and defines operational restrictions to ensure safe Airside Operations taking into account the available aerodrome facilities.

4. SAFEGUARDING PROCEDURES:

Safeguarding Procedures (SP) are instructions for relevant airport departments and airside operators to prepare ground services and facilities for low visibility operations in order that LVPs are implemented after all Safeguarding procedures are completed.

Duty Officer Tower/Tower Supervisor will initiate and co-coordinate with all the concerned agencies for completion of safeguarding procedures before implementation of Low Visibility Procedures.

4.1 Safeguarding Procedures shall be initiated when:

a. The Visibility/RVR is less than 1200M and visibility/RVR is forecast to deteriorate to 800M/550M or less; and/or
b. The cloud ceiling is less than 400FT and forecast to fall to 200FT or less.

4.2 Safeguarding procedures include:

a. Positioning of one CFT each at the two predetermined positions. PDPs for RWY 07/25 are near Gate no: 6 and near TWY K and for RWY 12/30 are on Grid B6 (near Adayar Bridge) and Gate no: 6.
b. Stopping of all maintenance works on the manoeuvring area as well as removal of all men and mobile equipment from the said area.
c. Inspecting Aeronautical ground lights (AGL) during the hour preceding implementation of LVP, and thereafter once every two-hour period.
d. Ensuring readiness of secondary power supply for change over time of maximum one second for AGL during LVP implementation. All Aerodrome Ground Lights may continue to operate on main power supply during Safe Guarding Procedures. Aerodrome Ground Lights (AGL) at Chennai Airport is supported by UPS to ensure change over time of less than one second. However, during any unserviceability of the UPS, whenever LVP is to be implemented as per para 6 below, the AGL shall be put on Standby Power Supply.
e. Apron Duty Manager (ADM) shall inspect the airfield and Aerodrome ground lights during the hour preceding implementation of LVP, and thereafter every subsequent two-hour period. These lighting inspections should be accorded priority and, if necessary, aircraft operations may have to be delayed.

5. FACILITIES SUPPORTING LVP:

The following Facilities are available at Chennai Airport for supporting Low Visibility Procedures (refer Table 1):

a. Runway Centreline Lights for RWY 07/25
b. Runway Edge lights for RWY 07/25 & 12/30
c. Runway End Lights for RWY 07/25 & 12/30
d. Runway Center line markings for RWY 07/25 & 12/30
e. Transmissometer for RWY 07/25

6. All Aerodrome Ground Lights at Chennai Airport are supported by UPS (Uninterrupted Power Supply) to ensure change over time of less than one second. However, during any unserviceability of the UPS when LVP in force, secondary power supply shall be switched on for supporting the AGL to ensure change over time of less than one second.

7. LOW VISIBILITY PROCEDURES:

7.1 Implementation of Low Visibility Procedures:

Duty MET Officer shall inform Tower Supervisor whenever Visibility/RVR reduces to 800/550M or below and/ or cloud

ceiling is less than 200FT. Tower Supervisor shall coordinate with AOCC, Fire Station and and Constant Current Regulator (CCR) to ensure that the safeguarding procedures are completed.

When Visibility/RVR falls below 800M/550M and or Cloud Ceiling less than 200FT and safeguarding procedures are completed, Tower supervisor will implement Low Visibility Procedures after prior approval from WSO. Anytime reported visibility is below 800M, then the 550M RVR criteria apply for this interpretation. Tower Supervisor shall ensure that "Low Visibility Procedure in Force" is being broadcast on ATIS.

Table 1: RVR/Visibility for Take-off (Commercial Transport Aeroplanes)	
Take-off RVR/Visibility	
Facilities	RVR/VIS (Refer Note 1) Cat A, B, C, & D
Adequate Visual reference (Day Only) (Refer Note 2)	500 M
Runway edge lights or Runway centre line markings (Refer Note 3)	400 M
Runway edge lights and Runway centre line markings (Refer Note 3)	300 M
Runway edge lights and Runway centre line lights	200 M
Runway edge lights, Runway centre line lights and relevant RVR information (Refer Note 4)	150 M
High Intensity Runway edge lights and Runway centre line lights (spacing 15m or less) and relevant RVR information (Refer Note 4)	125 M
High Intensity Runway edge lights and Runway centre line lights (spacing 15m or less), approved lateral guidance system and relevant RVR information (Refer Note 4)	75 M

Note 1 – The TDZ RVR/VIS may be assessed by the pilot. For take-off circumstances where TDZ RVR is inoperative or is determined by the pilot to be significantly in error (e.g. patchy fog obscuring Transmissometer but not the Runway, snow on Transmissometer causing erroneous readings), a pilot assessment may be made in lieu of RVR subject to the conditions laid down in CAR section 8, Series 'C', Part-I on All-weather Operations under para 17.1, 17.2, 17.3 & 17.4 (Pilot assessment of touchdown zone RVR is to be made only when the mid and roll out zone RVR are reported and both these values are not less than 200m).

Note 2 – Adequate Visual Reference means, that a pilot is able to continuously identify the take-off surface and maintain directional control.

Note 3 – For night operations at least runway edge lights or centre line lights and runway end lights are available.

Note 4 – The required RVR must be achieved for all relevant RVR reporting points (touchdown, mid-point and stop-end/ roll-out). The governing RVR shall be the lowest of the reported RVRs.

Note 5 - LVTO pertains to take off when the RVR is below 400 m and is applicable whenever the reported RVR in any zone (touch-down/mid/roll-out) RVR is below 400 m. The facilities and conditions specified in Table 1 will be as per the lowest RVR reported in any zone (e.g. if the RVR is 400/300/300 representing the three zones, then the 300 m will be the RVR for reckoning facilities and conditions of Table 1. If the RVR is 300/150/Not Reported, then 150 m will be the reckoning RVR and as the RVR is below 200 m all three RVRs are required).

7.2 Action by various agencies during LVP:

7.2.1 Met Office

a. The Duty Met Officer (DMO) shall issue "Outlook for Low visibility" whenever RVR is less than 1200 meters or visibility is forecast to deteriorate to 800 meters or less; and/or the cloud ceiling is 400 feet and forecast to fall to 200 Feet or less.

b. The DMO shall ensure that the Transmissometer is serviceable and direct reading of TDZ RVR observations are provided at control positions.

c. Whenever Transmissometer is unserviceable, or the LVTO is to be conducted on Runways not equipped with Transmissometer, DMO shall deploy suitable manpower to obtain Manual RVR.

d. Duty MET Officer shall keep Duty Officer Tower/Tower Supervisor informed of any change in Visibility/ RVR.

7.2.2 Apron Control

a. Restrict the number of the vehicles operating on the manoeuvring area to bare minimum and must have their obstruction lights ON during Low Visibility Operations.

b. Vehicles fitted with a serviceable R/T shall only be allowed to operate on the manoeuvring area. In case of extreme urgency, a vehicle without RT may be permitted provided it is escorted by a Follow Me Vehicle with a serviceable R/T and as far as practicable vehicles equipped with ADSB squitter only are allowed to operate on the movement area.

c. When LVP in force, Apron Duty Manager (ADM) shall ensure that secondary Power Supply is switched on, during the unserviceability of UPS, to ensure change over time of 1 second or less.

d. ADM should also ensure that markings are kept free of contamination and are sufficiently conspicuous to the aircraft during LVOP.

e. Any change to critical facilities and associated limitations shall be disseminated by ADM to ATC without delay.

f. ADM shall stop all work and withdraw all personnel working on the Runway strip and the manoeuvring area.

g. Apron Control shall ensure that all users of the Apron including airlines are informed of the imposition/ withdrawal of low visibility operational plan.

h. Provide all non-sked aircraft with Follow-Me service.

i. Provide follow-me service to all aircraft taxing in/out of Remote Aprons or at any time requested by pilot.

j. Apron Control shall ensure that the towing of aircraft is done under escort of "Follow Me" vehicles. Only one towing at a time shall be permitted.

7.2.3 Airport Rescue and Fire and Fighting services (ARFF)

a. The Officer in charge of Airport Rescue and Fire and Fighting services (ARFF) shall ensure that two CFT's are positioned at the pre-determined position after obtaining permission from ATC on R/T as defined in Aerodrome Emergency Plan.

b. ARFF must receive clearances from ATC prior to entry to any aircraft movement area other than the designated service roads.

7.2.4 Surface Movement Control

a. SMC shall not permit any ground run on the manoeuvring area except idle power run up on the stands, after coordination with AOCC.

b. Pushback shall be issued in such a manner that the aircraft movement on the apron is restricted to not more than three aircraft at a time.

c. When RWY Guard Lights are unserviceable, only one departure shall be allowed to taxi to the respective RWY holding point.

d. The following shall be included in ATIS: "**Low Visibility Procedures in Force**".

e. Surface Movement Controller (SMC) shall monitor the status of taxiway/runway lights and immediately advise the aircraft under its control of any unserviceability.

7.2.5 Tower Control

a. Duty Officer Tower/Tower Supervisor shall inform AOCC and ARFF of implementation of Low Visibility Procedures.

b. TWR shall permit departures only from the beginning of the Runway in use.

c. Duty Officer Tower/Tower Supervisor shall stop all Departures whenever RVR falls below the minima applicable and when Facilities, equipment and services necessary for Low Visibility Operations are degraded and/or the prevailing conditions are considered unsafe for such operations.

d. Tower Supervisor shall check Transmissometer displays.

e. Tower supervisor shall inform ADM/AOCC the implementation and cancellation of LVP.

7.2.6 Watch Supervisory Officer (WSO)

a. WSO shall initiate the implementation of Low Visibility Procedure and advise Tower Supervisor of the same.

b. WSO should consult Meteorological Office for forecast before terminating LVP.

c. WSO shall terminate LVP when Meteorological conditions improve and visibility /TDZ RVR is 800/550 meters or more and the cloud ceiling is 200 Feet or higher, and trend is for improvement for both runways.

7.2.7 Engineering Dept

a. In-Charge Electrical shall continuously monitor the main and Standby Power supply to ensure change over time of maximum one second for RWY Edge lights, RWY end lights and RWY CL lights during low visibility operations and report any unserviceability to Tower immediately.

7.2.8 Airlines

a. Pilots shall ensure and inform ATC that "RVR within Minima for LVTO" while requesting push back/start-up.

8. TERMINATION OF LOW VISIBILITY PROCEDURES:

a. When Visibility/RVR improves to 800/550M or more and cloud ceiling is 200FT or higher and trend is for improvement, Tower Supervisor/Duty Officer Tower would terminate operations of LVP. Tower shall obtain advice from Duty Met. Officer regarding improvement in weather conditions before the termination of LVP.

b. The Tower Supervisor will intimate Fire Station and AOCC regarding the termination of LVP operations.

c. On cancelling of LVP, following message shall be included in two subsequent ATIS broadcasts. "**LOW VISIBILITY PROCEDURES CANCELLED"**.

d. If SP are implemented and LVP are not subsequently implemented and the visibility/RVR improves and is more than 1200M and/or the cloud ceiling is 400FT or higher and both are forecast to remain above the required SP criteria, Tower Supervisor/Duty Officer Tower may cancel SP.

9. UNSERVICEABILITY OF AERONAUTICAL GROUND LIGHTS/ EQUIPMENT BEFORE/DURING IMPLEMENTATION OF LVP:

9.1 Low Visibility Procedures will not be implemented or will be restricted when any/all of the light/equipment mentioned in Para 5 above is/are un-serviceable or is/are not available as per Table 2 given below:

Table 2: Effect of Unserviceability of AGL on LVTO		
Aeronautical Ground Lighting Facility	Un-serviceability	Restrictions
Runway Edge lights/ Runway centreline lights	More than 15% of all lights are unserviceable	LVTO can continue if RWY Centre line lights and RWY end lights are available.
	Any two consecutive lights or more are unserviceable	
Standby Generator	Generator is unserviceable	LVP operations will be suspended unless the AGL is adequately supported by UPS
Runway End Lights	More than 15% of all lights are unserviceable	LVP operations will be suspended.
	Any two consecutive lights or more are unserviceable	
Runway Centreline lights	Any two consecutive lights or more are unserviceable.	LVTO only up to 300 M can be permitted, subject to the remaining facilities being serviceable with realtime TDZ RVR availability
	More than 15% of all lights are unserviceable	
Real time TDZ/MID/END RVR.	Non-availability of Real time MID/ END RVR.	LVTO only up to 200 M can be permitted, subject to remaining facilities being serviceable with real-time TDZ RVR availability

In case more than one facility is not available, higher of the RVR requirement of the individual cases should be considered as RVR for permitting LVTO.

9.2 A lighting system is deemed to be on outage when:

a. In the case of a lighting system comprising of 6 to 13 lights (e.g. threshold lights) more than 2 lights become unserviceable, or two adjacent lights become unserviceable.

b. In the case of a lighting system comprising more than 13 lights, more than 15% of the lights become unserviceable, or two adjacent lights become unserviceable.

10. GENERAL ACTIONS BY OTHER AGENCIES (AIRLINES, REFUELLING COMPANIES, CATERING AGENCIES, ETC):

a. Every year before commencement of monsoon/winter season, a meeting will be held by GM (GFS), to inform all airlines and agencies operating at airport about their roles/responsibilities and create awareness to ensure cooperation for safe airport operations during periods of low visibility.

b. All the agencies shall ensure that staff and drivers are suitably trained for operations during Low Visibility Operations.

c. All agencies operating in the operational area shall ensure that only those vehicles that are absolutely essential for aircraft operations are used in the operational area during periods of low visibility. The drivers of these vehicles should keep a look out for taxiing aircraft and other vehicles to prevent accidents.

d. All agencies shall ensure that vehicle with a serviceable R/T only and as far as practicable vehicles equipped with ADS-B squitter only are allowed to operate on the movement area.

e. All the vehicles must have their obstruction lights “ON” during Low Visibility Procedures operations.

f. A refresher program for ATCO’s and personnel responsible for airside operations shall be conducted every year.

g. All instructions/sign boards provided for vehicular movement area/service roads, must be followed while operating in the operational area

RADIO COMMUNICATION FAILURE PROCEDURES

CHENNAI INTERNATIONAL AIRPORT, CHENNAI (VOMM)

1.INTRODUCTION

The Radio Communication Failure procedures established herein are based on the provisions of ICAO ANNEX 2, DOC 4444 PANS ATM and DGCA CAR Section 9, Series C, Part 1. The objective is to standardize the actions to be taken by the Pilots of Arriving and Departing Aircraft at Chennai Airport experiencing Radio Communication Failure.

2.SCOPE:

The procedures in this document shall be applicable to the pilots of Arriving and Departing Aircraft experiencing Radio Communication Failure at Chennai Airport.

3.GENERAL:

3.1RCF can be of four types:

a)Aircraft receiver failure but transmitter functioning.

b)Aircraft receiver and transmitter both failed.

c)Aircraft transmitter failure but receiver functioning.

d)Ground communication failure

3.2In case the cockpit crew is able to receive instructions from ATC on VHF, ATC will issue appropriate instruction to guide the crew and the crew shall comply with the instructions issued by ATC. [Standard ATC instructions in such cases would instruct crew to squawk ident, execute specific turn manoeuvres, etc.to confirm receipt of ATC transmissions]

3.3In case the cockpit crew is unable to receive instructions from ATC on VHF, the under mentioned procedures given in Para 5 & 6 shall be followed.

3.4In case of RCF, whether complete or partial, cockpit crew must make blind transmission of all actions on RT. This will ensure ATC has information of the flight manoeuvres.

3.5In all cases of RCF, the cockpit crew will set SQUAWK 7600 at the earliest. [Note: This Requirement of Setting transponder to Mode A/C code 7600 in no way imposes any restriction on the pilot’s decision to set transponder to Mode A/C code 7500 or 7700, whenever required]

3.6Unless otherwise required, aircraft shall avoid all restricted areas. In case an aircraft is observed approaching any of such airspace, the concerned ATS unit shall inform the respective units accordingly. Pilot shall not overfly VOP193 (Prohibited Area) Kalapakkam (FL100/GND), which is situated 25Nm south of Chennai and active danger areas.

3.7Pilots should ensure that they remain at or above the minimum sector altitude. If the aircraft is below the minimum sector altitude/ minimum flight altitude, pilots shall immediately climb to the minimum sector/flight altitude

4.ASSIGNED RUNWAY AND ITS AVAILABILITY FOR RCF AIRCRAFT

4.1In case of arriving aircraft, when Runway for landing has already been advised to the aircraft by ATC, such runway shall be considered as assigned runway. In case, arriving aircraft has not been assigned any Runway, Runway 07 shall be considered as assigned Runway for such arrival, except during the notified periods of maintenance/closure of Runway 07, Runway 12 shall be considered as assigned runway

.

4.2To assist the pilot of aircraft experiencing RCF, Runway lights, PAPI & Approach lights in 'SWITCHED ON' mode shall indicate the availability of runway for landing
4.3In case of departures from Chennai returning on account of RCF, assigned runway for landing is as given below, except during the notified periods of maintenance/closure of runway wherein the departure runway of such aircraft shall be considered as assigned runway
4.3.1For RWY 07 & RWY 12, Assigned runway for landing will be RWY 07,
4.3.2For RWY 25 & RWY 30, Assigned runway for landing will be RWY 25,
4.3.3Airlines shall file take-off alternate, when main runway 07/25 is under NOTAM and unable to land on the runway of departure.
4.3.4Aircraft shall use MMV VOR for RWY 07/25.
4.3.5Aircraft shall use CNI VOR for RWY 12/30, when RWY 07/25 is not available.
5.PROCEDURE FOR ARRIVALS:
5.1GENERAL:
5.1.1 If in VMC, continue to fly in VMC and observe for the Light signals from Chennai Tower and land.
5.1.2 If in IMC or when the pilot of an IFR flight considers it inadvisable to complete the flight in accordance with para 5.1.1 above, the pilot shall follow as described below

PHASE OF ARRIVAL	PROCEDURES BY PILOT	ACTION BY ATCOS (on observing A7600 for 2 Min)
Any phase except final approach	a)In case of RNAV arrivals which has received arrival clearance, shall follow the RNAV STAR with all lateral and vertical restrictions and at the end of the STAR shall make a convenient turn to intercept the LLZ/ final approach track. b)In case of RNAV arrivals which has not received arrival clearance, shall proceed to follow the RNAV STAR appropriate to the route with all lateral and vertical restrictions and at the end of the STAR shall make a convenient turn to intercept the LLZ/final approach track. (Ref section 4 on RWY assignment & availability) c)Arrivals following STAR shall follow the RCF procedures when specified in the appropriate published STAR in line with a) or b) above. d)Non-RNAV arrivals shall proceed to VOR maintaining last assigned level or FL70 whichever is higher. If descend clearance has not been received, three minutes after setting transponder to Mode A/C Code 7600, shall commence descend to F70 and proceed to VOR. On joining VOR hold, descend in the hold to 3000 feet and execute: i.RWY 07/25: ILS Approach. In case ILS is not available, then execute VOR approach as per RWY in Use. ii.RWY 12/30: proceed to CNI for VOR approach. (Ref section 4 on RWY assignment & availability)	ATC will endeavour to provide larger than the prescribed separation with respect to RCF aircraft from other aircraft as appropriate
If landing clearance not received	If aircraft on final approach has not received landing clearance, Squawk A7600 and a)continue approach to land if visual with the runway or b)carry out missed approach as published. Proceed for next approach from VOR (MMV or CNI as the case be) for the same runway.	ATC will keep the runway approach lights and associated AGL in switched "ON "mode for the next approach. ILS as applicable will be kept on switched "ON" mode

Radar Vectored Arrival	Aircraft shall a)maintain last assigned level and heading for 3 minutes after detecting RCF or Selecting MODE A/C 7600 whichever is later and b)shall proceed to VOR maintaining last assigned level or FL70 whichever is higher c)On joining VOR hold, descend in the hold to 3000 feet and execute i.RWY 07/25: ILS Approach. In case ILS is not available, then execute VOR approach as per RWY in Use or RNP approach RWY 07/25 ii.RWY 12/30: proceed to CNI for VOR approach or RNP approach RWY 12/30.(Ref section 4 on RWY assignment & availability	

5.1.3In case strong tail wind conditions are encountered during approach for the assigned Runway, the aircraft experiencing RCF will carry out a missed approach on assigned runway and after following complete missed approach procedure for such approach, aircraft will carry out approach procedure published for the opposite runway
5.2Total Radio Communication Failure for aircraft carrying out missed approach:
5.2.1.Set aircraft transponder to Mode A Code7600 and proceed direct to the appropriate missed approach holding point (VOR) and join hold at 3000 ft as published.
5.2.2.The pilot shall climb and maintain 4000 ft in the holding pattern, complete one hold and commence an appropriate approach procedure.
5.3Partial radio communication failure for Arriving Aircraft\
5.3.1Aircraft Unable to Receive: Pilots shall adopt the total radio communication failure procedures specified in para 5.2 above.
5.3.2Aircraft Able to Receive: Following verification that aircraft is able to receive ground transmissions by squawk ident, ATC will continue to issue and repeat instructions and/or clearances to the pilot

6.PROCEDURE FOR DEPARTURES:

6.1ACTION BY PILOT IN CASE OF RCF AT DIFFERENT STAGES OF FLIGHT AFTER DEPARTURE

6.1.1Departure Intending to Continue to Destination
6.1.1.1A departing aircraft experiencing radio communication failure and intending to continue to its filed plan destination shall:
Three minutes after setting Mode A/C code 7600 or reaching FL60 or cleared Flight level (if higher than FL60) whichever is later:
a)if following SID, continue on SID to join ATS route and climb to filed Flight level and continue as per the filed flight plan.
b)if following Radar heading or have been directed by ATC to proceed offset using RNAV without a specified limit, proceed in the most direct manner possible to re- join the current flight plan route no later than the next significant point, maintaining cleared flight level/FL60 whichever is higher. After joining ATS route, climb to filed Flight level and continue as per the filed flight plan.
6.1.2Departure Intending to Land Back at Chennai:
6.1.2.1A departing aircraft, experiencing radio communication failure and intending to land back at Chennai Airport in accordance with provisions of DGCA CAR Section 9 Series C part I, shall:
a)Continue on assigned SID or radar heading and set Mode A/C code 7600.
b)Climb/ stop climb at a level according to the table given below:

Phase of flight	Actions to be followed by pilot
At or below FL70	Maintain/Climb to FL 70
Above FL70, but below FL230	Maintain/ Climb to: The last cleared and acknowledged level or FL230/ FL220, (appropriate to the direction of the flight) or flight planned level whichever is lower
At or above FL230	Stop climb at the next level appropriate to the direction of the flight.

c)On reaching a suitable level according to Para 6.1.2.1 (b) above, or three minutes after setting Mode A/C code 7600, whichever is later, make shorter arc to come over VOR.

•

If fuel jettisoning is not necessary:

-Within 50 NM climb/descend/ maintain FL 70 and join VOR hold.

•

If fuel jettisoning is necessary:

-Within 50 NM climb/descend/maintain FL 100 and join VOR hold

-After one hold overhead, proceed to Fuel dumping area (as described in Para 6.1.3)

-After jettisoning the fuel, proceed to join VOR hold at FL 100.

•Descend to 3000ft in the hold, leave VOR for the laid down procedure and continue descend as per the procedure for assigned runway. [refer para 4 for assigned runway and landing clearance and as specified in para 5.1.2]

6.1.2.2Aircraft departing to the south of Chennai shall avoid VOR183A. Hence aircraft

a)from RWY 07/12 shall turn left to proceed to VOR

b)from RWY 25/30 shall turn right to proceed to VOR

6.1.3Fuel dumping area at Chennai is earmarked as below:

a)An area bounded by MMV R030 and R060

b)beyond 15NM, between FL075-FL150

VOMM AD 2.23 ADDITIONAL INFORMATION

1. Radio Altimeter operating area is established in the pre THR area of RWY 25. The dimension of area is 30M X 120M.
2. New ATC Tower coordinates 125916.112N 0801011.965E top elevation 55.6M/182.42FT AMSL located 890.8M, 207.32 DEG from ARP penetrates the Obstacle Limitation Surfaces.

3. **RUNWAY MAINTENANCE WORKS:**

3.1 **Activity: Rubber Deposit Removal work**

Days: All Tuesdays and Saturdays

Timings: From 0830 UTC to 0930 UTC.

Location: 765 m from the beginning of RWY 25 including intersection of RWY 07/25 and RWY 12/30.

3.1.1 Details of facilities closed, associated restrictions and precautions during works:

i. During the work period arrivals and departures on RWY 07 NOT permitted.

ii. RWY 25 available only for departures from TWY C intersection. Arrivals on RWY 25 NOT permitted.

iii. During work period the revised declared distances of RWY 25 are as follows:

RWY	TORA	TODA	ASDA	LDA
25	2735M	2940M	2718M	----

iv. In case of requirements / emergency, full length of RWY 07/25 will be made available within 15 minutes for Code E aircraft operations and RWY 12/30 will be made available within 10 minutes up to Code C Aircraft operations.

v. Runway intersection and TWY J NOT available for operations.

vi. RWY 12/30 available for Taxiing and Helicopter operations only.

vii. Aircraft parked on aircraft stand 01 to 06 shall pushback facing only west during work period.

viii. Refer Aerodrome Chart - Rubber Deposit Removal Work

3.2 **Activity: Runway Maintenance Work**

Days: All Tuesdays and Saturdays

Timings: From 0930 UTC to 1130 UTC.

Location: On RWY 07/25 excluding intersection of runway.

3.2.1 Details of facilities closed, associated restrictions and precautions during works:

i. RWY 07/25 NOT be available for operations.

ii. Junction of TWY D and TWY Q available for crossing RWY 07/25.

iii. Junction of TWY M and TWY P available for crossing RWY 07/25

iv. During VIP movement and adverse weather conditions work will not be carried out and RWY 07/25 will be available for operations.

v. Work will be suspended when visibility/trend forecast for visibility indicates below minima for operations on RWY 12/30.

vi. In case of emergency RWY 07/25 will be made available on 15 minute prior notice.

vii. During this work period all aircraft are advised to carry sufficient fuel to avoid diversion.

viii. Refer Aerodrome Chart- Runway Maintenance Work.

4. NOTE:

a. Refueling at stands no. 26, 27, 27L, 27R, 28, 28L, 28R, 29, 29L, 29R is carried out through fuel bowsers.

b. VDGS available for primary and secondary stands

c. Details of Taxiway D1

Location : South of TWY B connecting TWY H

Length : 150.93M

Width of Shoulder: 10.5M

Visual Aid : TWY Edge lights, all mandatory markings and signage as per CAR

d. Details of RET Z

Location : RET Z at a distance of 1831 M from RWY 07 connecting TWY B

Width of Shoulder: 10.5M

Visual Aid : RETIL, TWY Edge Lights, all mandatory markings and signage as per CAR

5.

a) Bird activity in and around the airfield due to grass cutting operations along thebasic strip of RWY. Pilot to exercise caution.

b) Laser activity on the approach path of RWY 07 & 25. Pilot to exercise caution

VOMM AD 2.24 CHARTS RELATED TO AN AERODROME

1. Aerodrome Chart (14 MAY 2026)
2. Aerodrome Chart (Details of RWY 12/30)
3. Aerodrome Chart (Hot Spot) (14 MAY 2026)
4. Aircraft Parking/Docking Chart (Apron - I, II, III) (19 March 2026)
5. Aircraft Parking/Docking Chart -Apron A (19 March 2026)
6. Aircraft Parking/Docking Chart Apron B-T (19 March 2026)
7. Aerodrome Obstacle Chart Type – A (Operating Limitations) RWY 07 (02 October 2025)
8. Aerodrome Obstacle Chart Type – A (Operating Limitations) RWY 25 (02 October 2025)
9. Aerodrome Obstacle Chart Type - A (Operating Limitations) RWY 12 (02 October 2025)
10. Aerodrome Obstacle Chart Type - A (Operating Limitations) RWY 30 (02 October 2025)
11. Aerodrome Obstacle Chart - Type B (02 October 2025)
12. VOR Procedure (DME Required) RWY 12 (9 JULY 2026)
13. VOR Procedure (DME Required) RWY 30 (9JULY 2026)
14. RNP Y Procedure RWY 07 (17 April 2025)
15. Coding Table - RNP Y Procedure RWY 07 (17 April 2025)
16. RNP Y Procedure RWY 25 (17 April 2025)
17. Coding Table - RNP Y Procedure RWY 25 (17 April 2025)
18. RNP Y Procedure RWY 12 (17 April 2025)
19. Coding Table - RNP Y Procedure RWY 12 (17 April 2025)
20. RNP Y Procedure RWY 30 (27 November 2025)
21. Coding Table - RNP Y Procedure RWY 30 (27 November 2025)
22. Aerodrome Chart - Rubber Deposit Removal Work Area
23. Aerodrome Chart - Runway Maintenance Work Area
24. Take Off Run Available from Intersection RWY 07(14 MAY 2026)

25. Take Off Run Available from Intersection RWY 25(14 MAY 2026)
26. Take Off Run Available from Intersection RWY 12(14 MAY 2026)
27. Take Off Run Available from Intersection RWY 30(14 MAY 2026)
28. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 07, DOKET 2A,VATMO 2A (9 JULY 2026)
29. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 07, INUVTI 2A, SAGOR 2A, AKMIL 2A (9 JULY 2026)
30. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 07, DOLKI 2A, DORAM 2A, KAMGU 2A (9 JULY 2026)
31. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 07, GULAM 2A, XIVIL 2A (9 JULY 2026)
32. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 07, AVNOS 2A, GIRNA 2A (9 JULY 2026)
33. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 07, AVMOR 2A, IDASO 2A (9 JULY 2026)
34. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 07, TABULAR DESCRIPTION-1 (9 JULY 2026)
35. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 07, TABULAR DESCRIPTION-2 (9 JULY 2026)
36. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 07, TABULAR DESCRIPTION-3 (9 JULY 2026)
37. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 25, DOKET 2B, VATMO 2B (9 JULY 2026)
38. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 25, DOLKI 2B, DORAM 2B, KAMGU 2B (9 JULY 2026)
39. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 25, GULAM 2B, XIVIL 2B (9 JULY 2026)
40. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 25, AVMOR 2B, IDASO 2B (9 JULY 2026)
41. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 25, TEGUT 2B, KKP 2B (9 JULY 2026)
42. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 25, TABULAR DESCRIPTION-1 (9 JULY 2026)
43. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 25, TABULAR DESCRIPTION-2 (9 JULY 2026)
44. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 25, TABULAR DESCRIPTION-3 (9 JULY 2026)
45. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 12, DOKET 2C, VATMO 2C (9 JULY 2026)
46. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 12, NUVTI 2C, SAGOR 2C, AKMIL 2C (9 JULY 2026)
47. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 12, DOLKI 2C, DORAM 2C, KAMGU 2C (9 JULY 2026)
48. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 12, GULAM 2C, XIVIL 2C (9 JULY 2026)
49. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 12, AVNOS 2C, GIRNA 2C (9 JULY 2026)
50. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 12, AVMOR 2C, IDASO 2C (9 JULY 2026)
51. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 12, TABULAR DESCRIPTION-1 (9 JULY 2026)
52. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 12, TABULAR DESCRIPTION-2 (9 JULY 2026)
53. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 12, TABULAR DESCRIPTION-3 (9 JULY 2026)
54. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 30, DOKET 2D,VATMO 2D (9 JULY 2026)
55. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 30, DOLKI 2D, DORAM 2D, KAMGU 2D (9 JULY 2026)
56. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 30, GULAM 2D, XIVIL2D (9 JULY 2026)
57. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 30, AVMOR 2D, IDASO 2D (9 JULY 2026)
58. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 30, TEGUT 2D, KKP 2D (9 JULY 2026)

59. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 30, TABULAR DESCRIPTION-1 (9 JULY 2026)
60. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 30, TABULAR DESCRIPTION-2 (9 JULY 2026)
61. Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 30, TABULAR DESCRIPTION-3 (9 JULY 2026)
62. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 07, TULNA 2E, ADKAL 2E, TEGUT 2E, NOPES 2E (9 JULY 2026)
63. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 07, OPAMO 2E, ANIRO 2E (9 JULY 2026)
64. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 07, AVMOR 2E (9 JULY 2026)
65. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 07, BOPRI 2E, RINTO 2E, GUNRI 2E, GURAS 2E, DOKET 2E (9 JULY 2026)
66. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 07, TABULAR DESCRIPTION-1(9 JULY 2026)
67. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 07, TABULAR DESCRIPTION-2 (9 JULY 2026)
68. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 07, TABULAR DESCRIPTION-3(9 JULY 2026)
69. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 25, AKMIL 2F(9 JULY 2026)
70. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 25, TULNA 2F, ADKAL 2F, TEGUT 2F, NOPES 2F(9 JULY 2026)
71. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 25, OPAMO 2F, ANIRO 2F(9 JULY 2026)
72. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 25, AVMOR 2F(9 JULY 2026)
73. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 25, BOPRI 2F, RINTO 2F,GUNRI 2F, GURAS 2F,DOKET 2F(9 JULY 2026)
74. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 25, TABULAR DESCRIPTION-1(9 JULY 2026)
75. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 25, TABULAR DESCRIPTION-2(9 JULY 2026)
76. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 25, TABULAR DESCRIPTION-3(9 JULY 2026)
77. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 12, TULNA 2G, ADKAL 2G, TEGUT 2G, NOPES 2G(9 JULY 2026)
78. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 12, OPAMO 2G, ANIRO 2G(9 JULY 2026)
79. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 12, AVMOR 2G(9 JULY 2026)
80. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 12, BOPRI 2G, RINTO 2G, GUNRI 2G, GURAS 2G, DOKET 2G(9 JULY 2026)
81. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 12, TABULAR DESCRIPTION-1(9 JULY 2026)
82. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 12, TABULAR DESCRIPTION-2(6 AUG 2026)
83. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 12, TABULAR DESCRIPTION-3(9 JULY 2026)
84. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 30, AKMIL 2H(9 JULY 2026)
85. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 30, TULNA 2H, ADKAL 2H, TEGUT 2H, NOPES 2H(9 JULY 2026)
86. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 30, OPAMO 2H, ANIRO 2H(9 JULY 2026)
87. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 30, AVMOR 2H(9 JULY 2026)
88. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 30, BOPRI 2H, RINTO 2H,GUNRI 2H, GURAS 2H,DOKET 2H(9 JULY 2026)
89. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 30, TABULAR DESCRIPTION-1(9 JULY 2026)

90. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 30, TABULAR DESCRIPTION-2(9 JULY 2026)

91. Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 30, TABULAR DESCRIPTION-3(9 JULY 2026)

VOMM AD 2.25 VISUAL SEGMENT SURFACE (VSS) PENETRATION

Procedure	Procedure Minima	VSS Penetration
1	2	3